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(54) **STOPPER**

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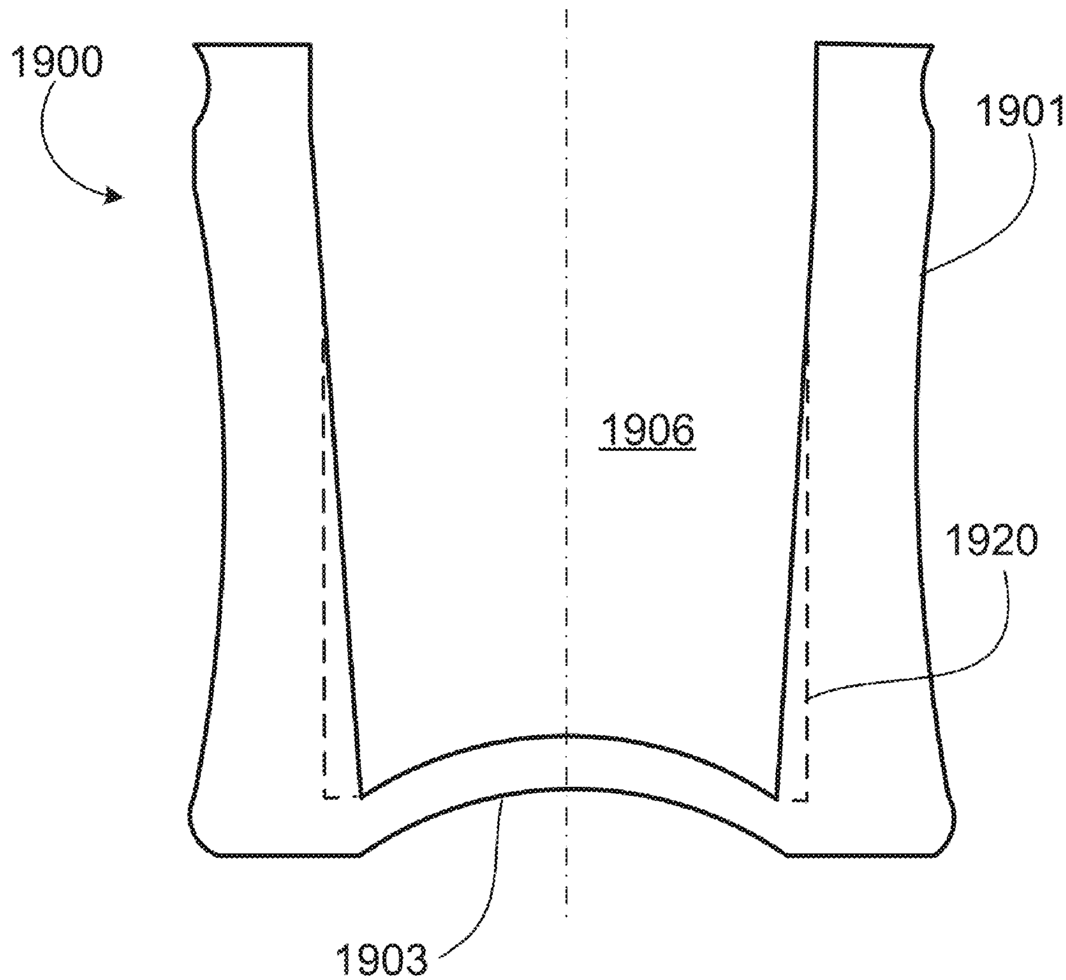
(57) **ABSTRACT**

A stopper for use in a cartridge or syringe of a medical device is configured to be disposed within a container closure system. The stopper comprises a shell comprising a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell. The open end defines a cavity and the sidewalls define an exterior surface sized and shaped to fit inside the container closure system. An insert is configured to be inserted into the cavity, receive a force from a plunger rod, and distribute the force to the shell in order to advance the shell into the container closure system. The closed end of the shell in the cavity defining a convex surface configured to be contacted and deflected by the insert upon insertion into the cavity includes an elastic and/or plastic deformable material.



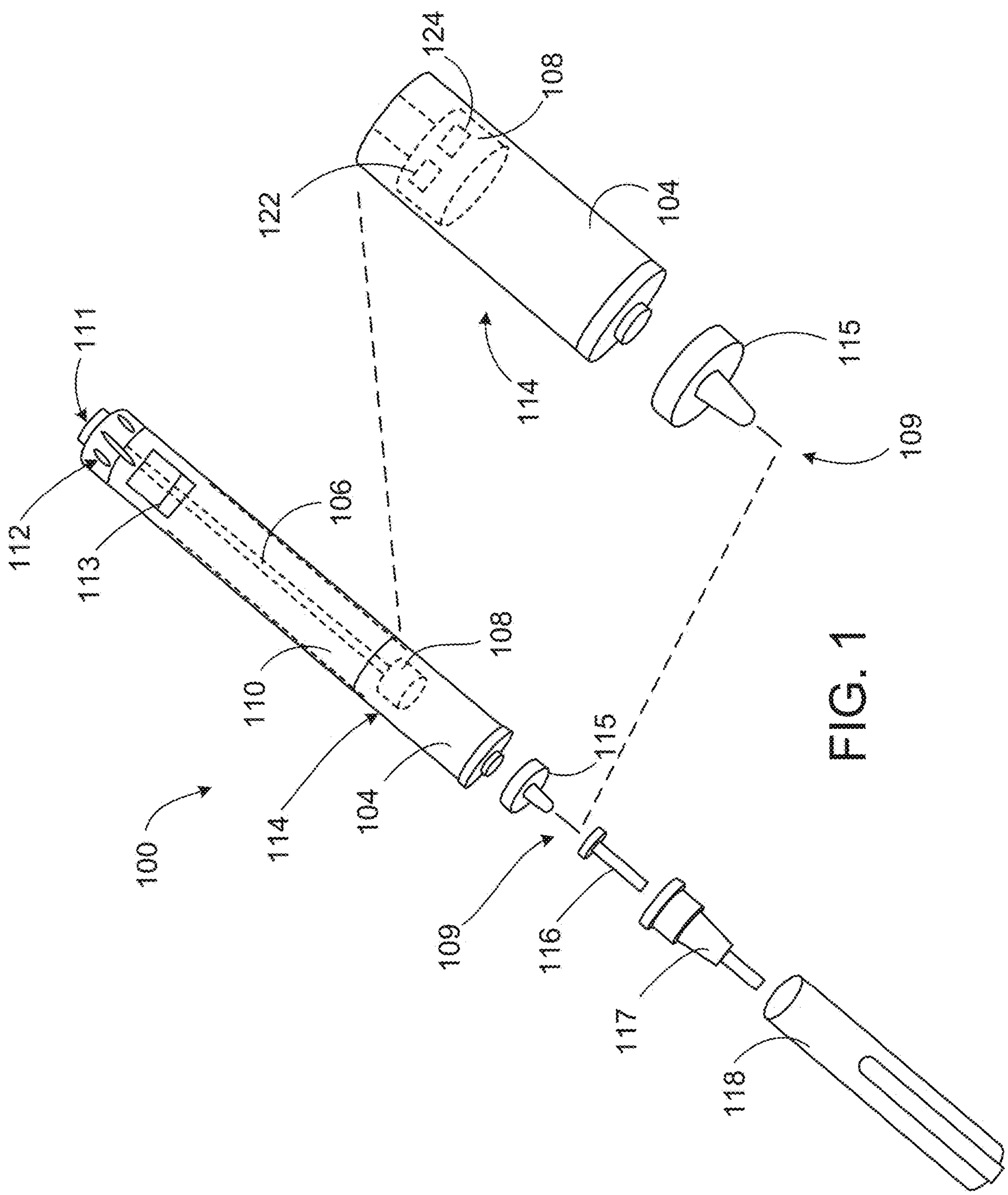

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FIG. 1

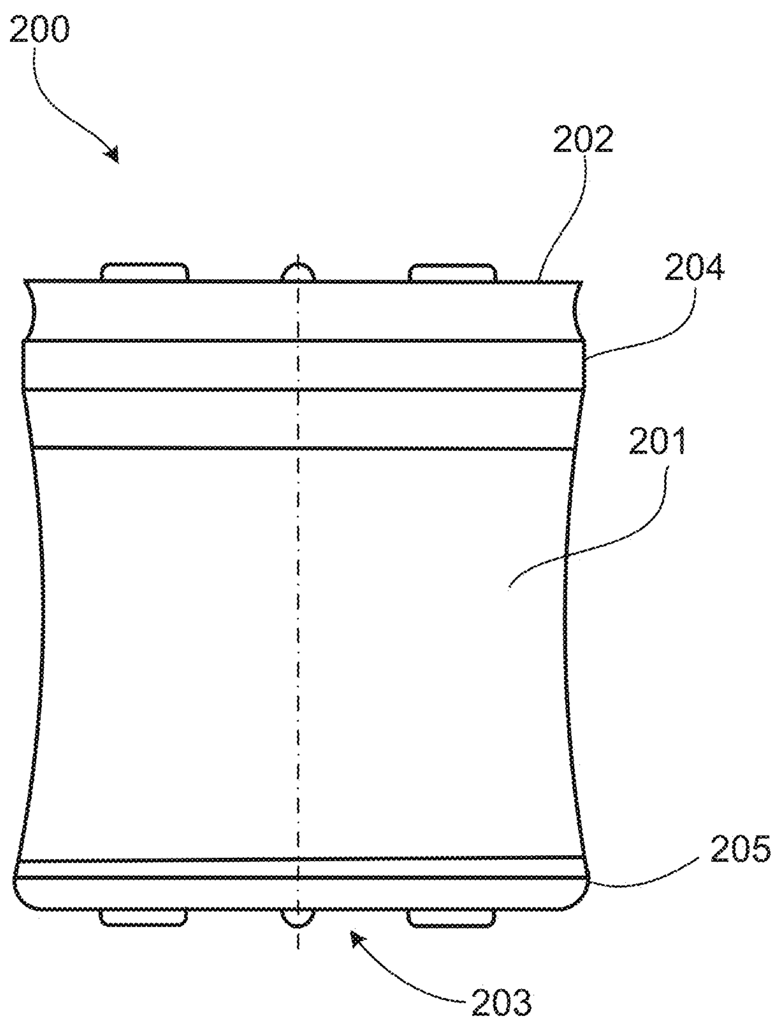

FIG. 2A

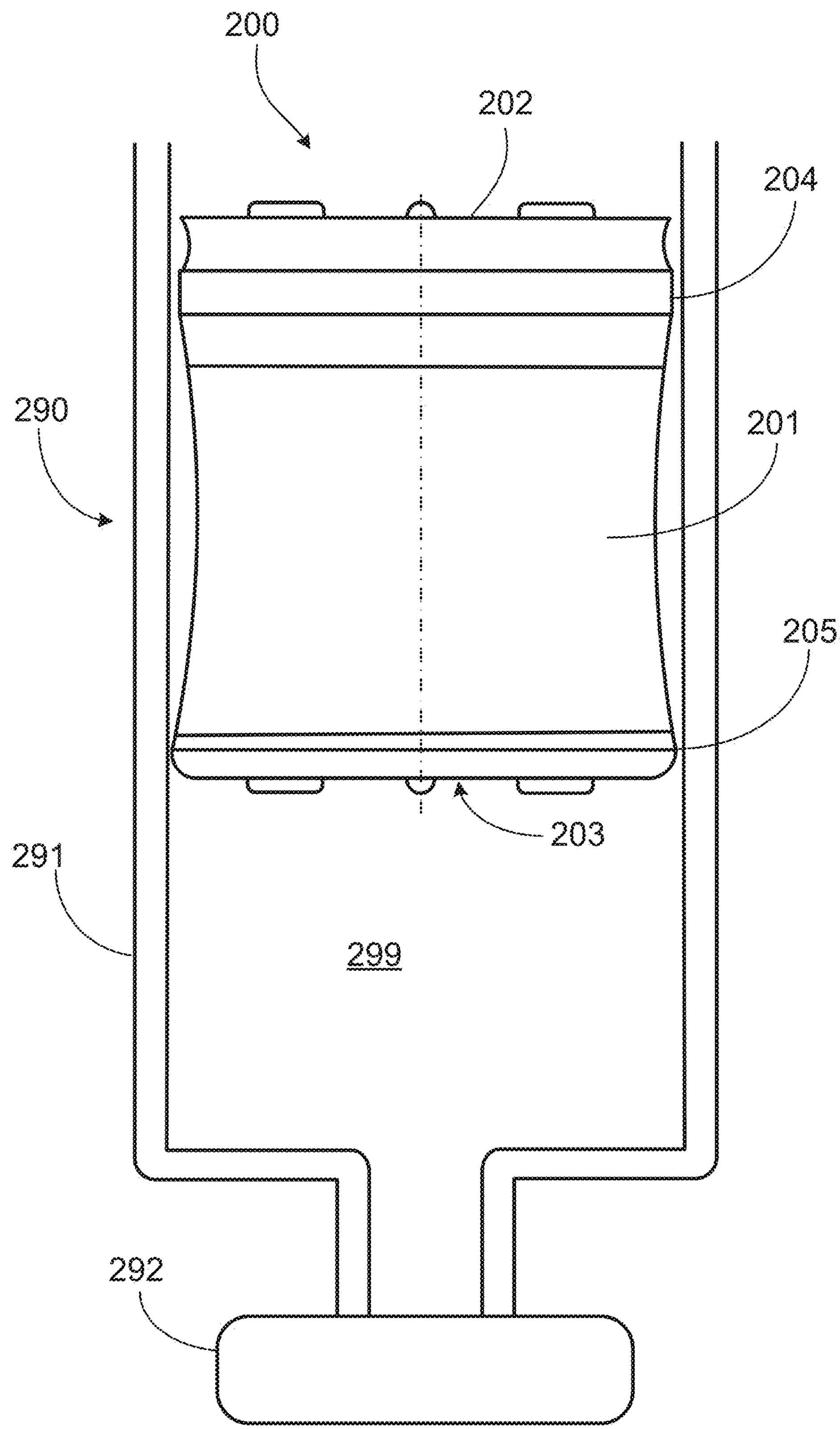

FIG. 2B

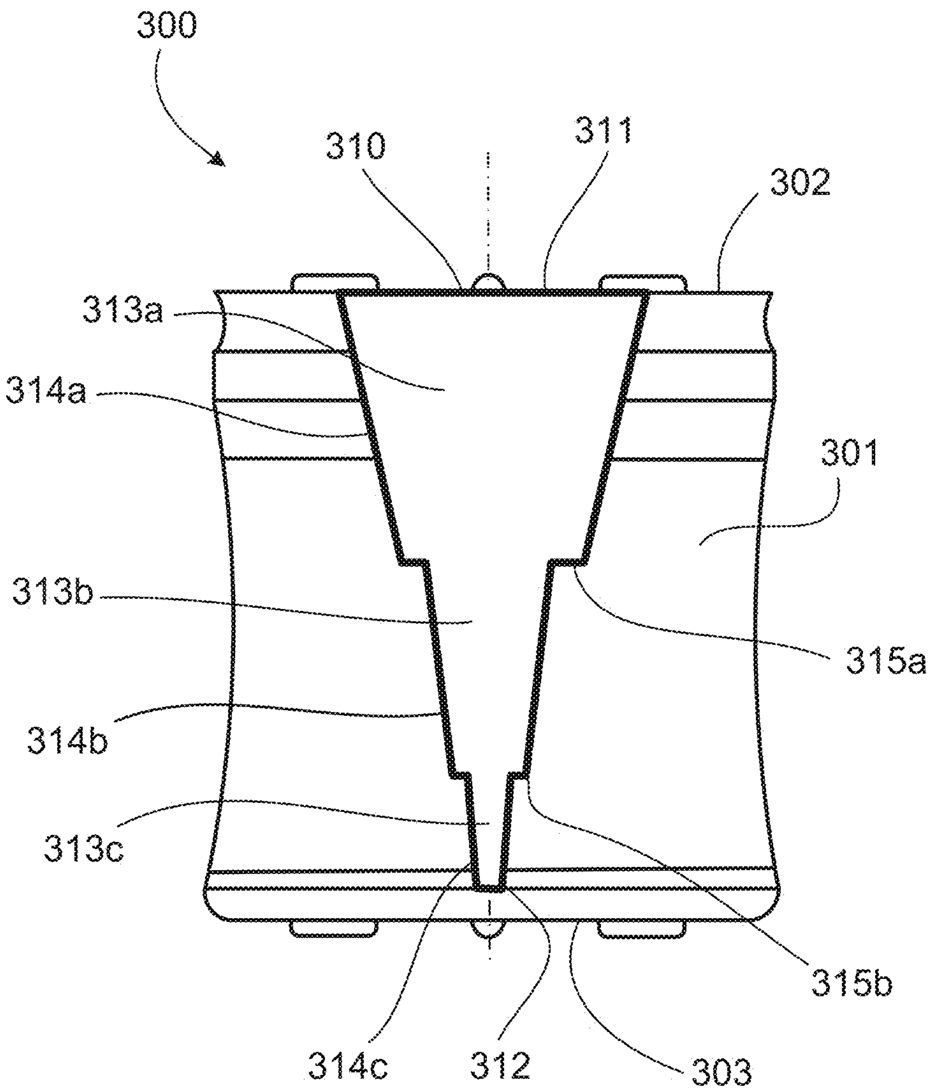
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FIG. 3

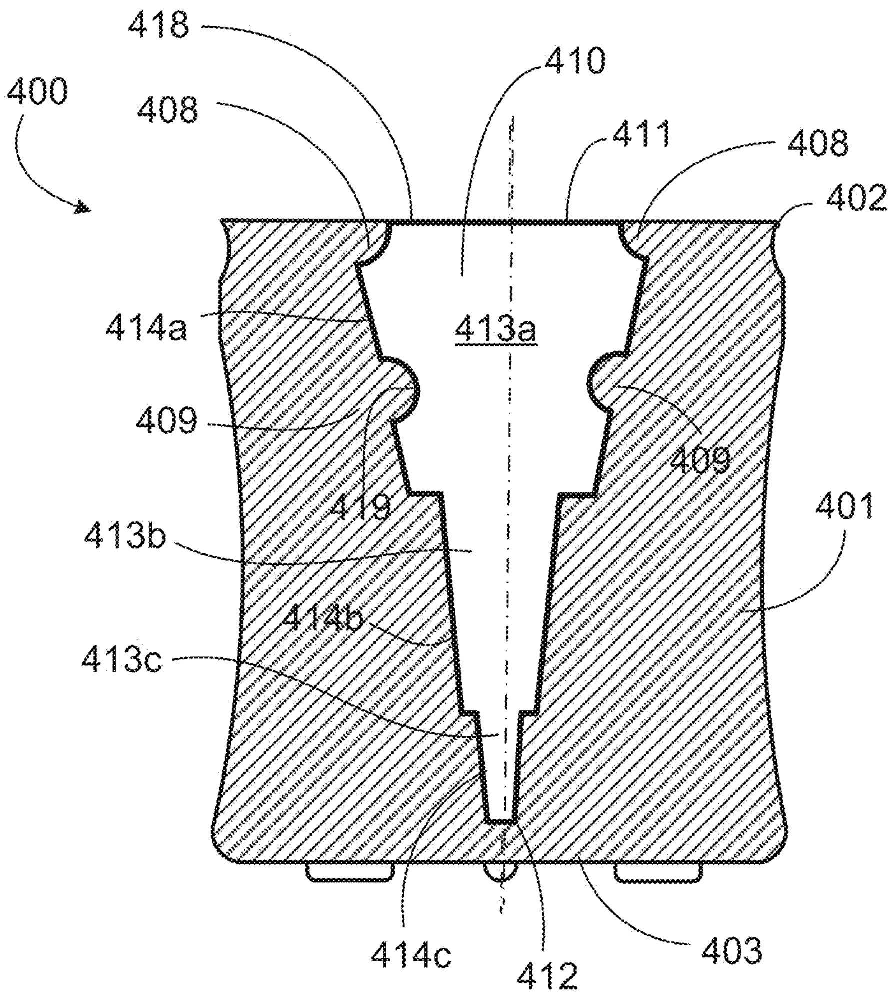
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FIG. 4

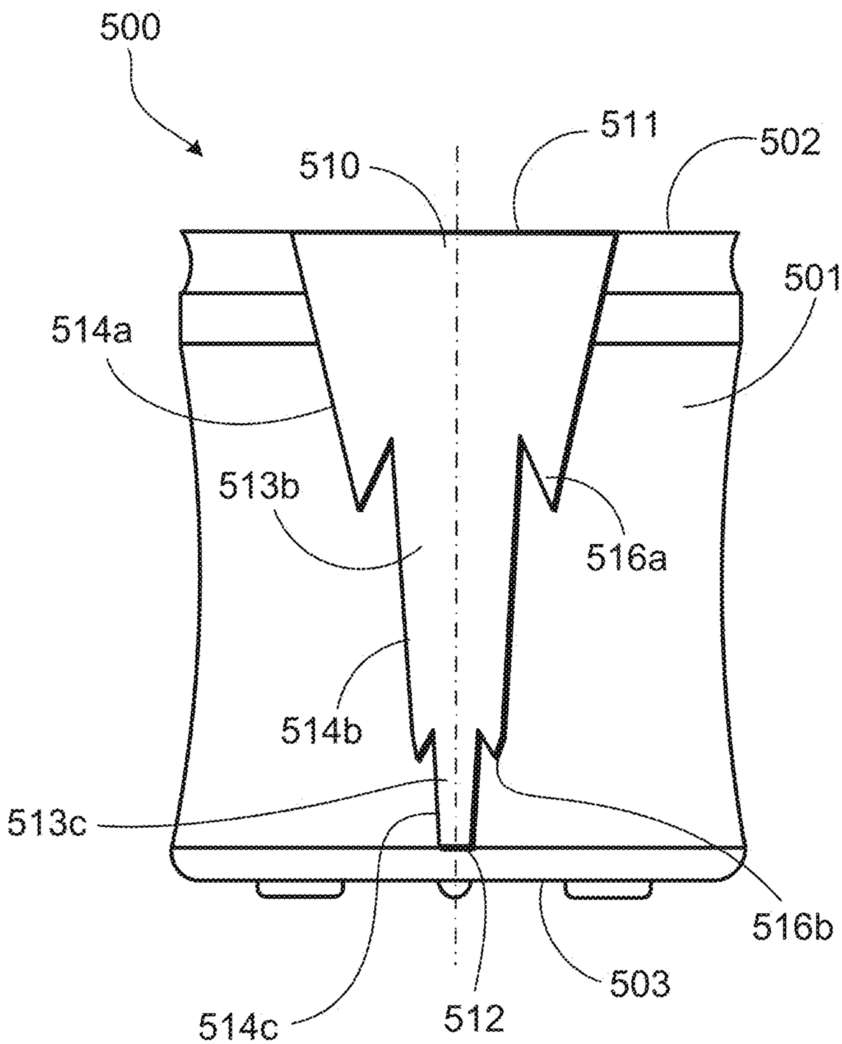


FIG. 5

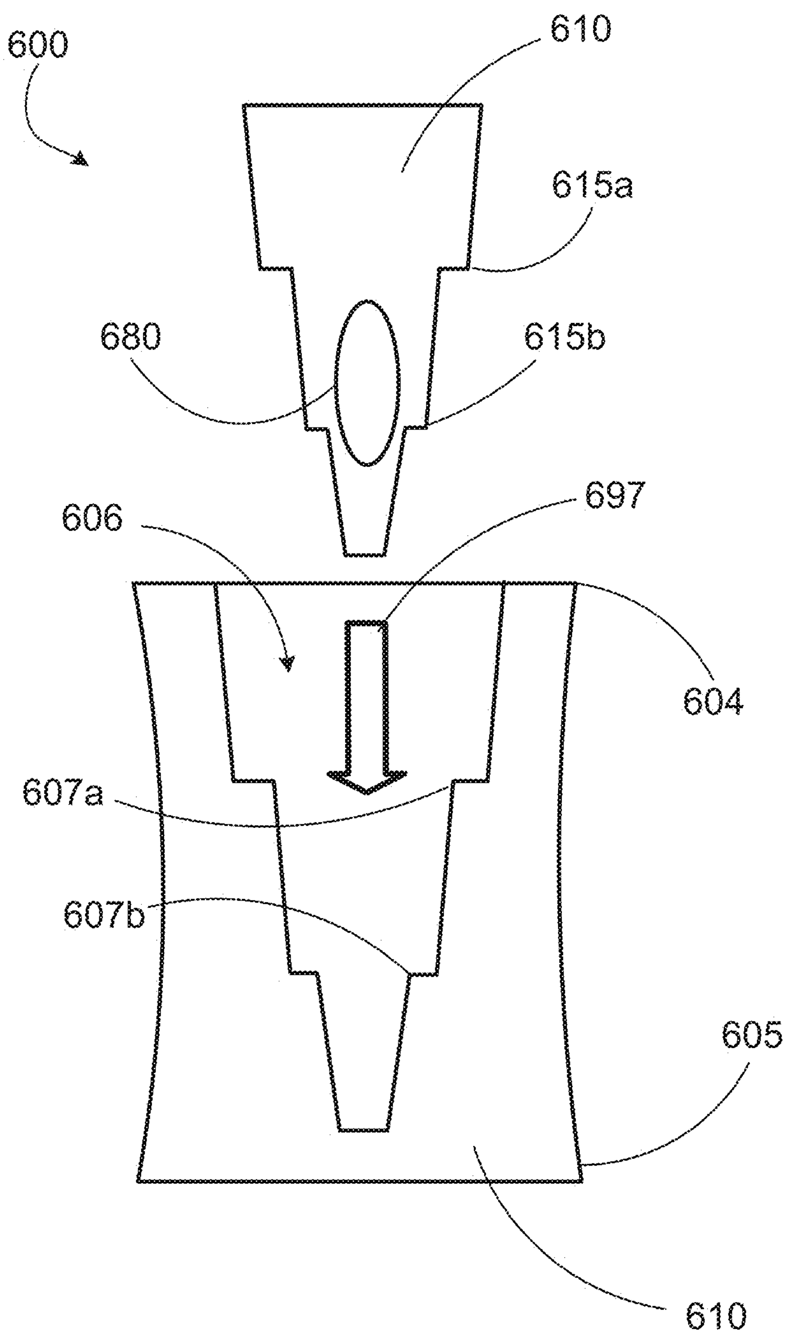


FIG. 6A

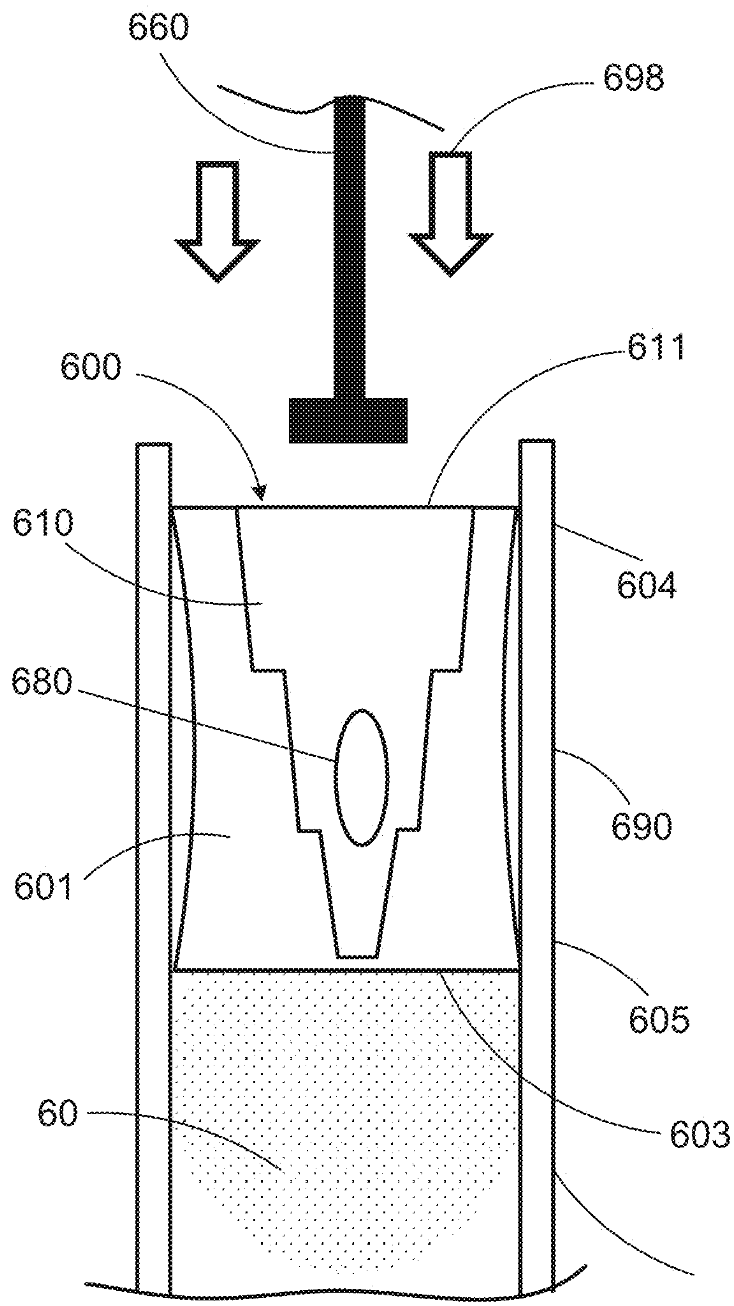
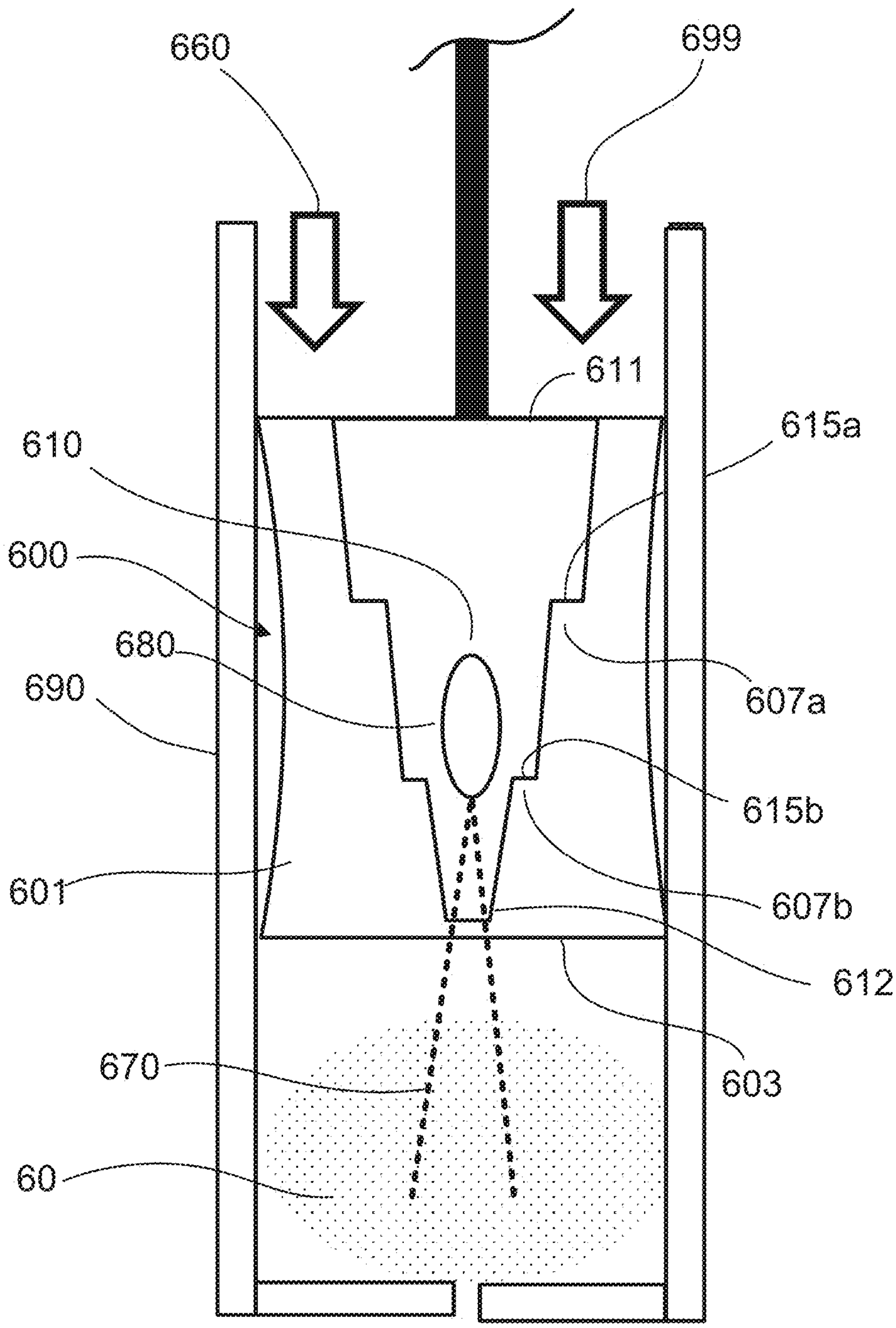


FIG. 6B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607b
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603
60

FIG. 6C

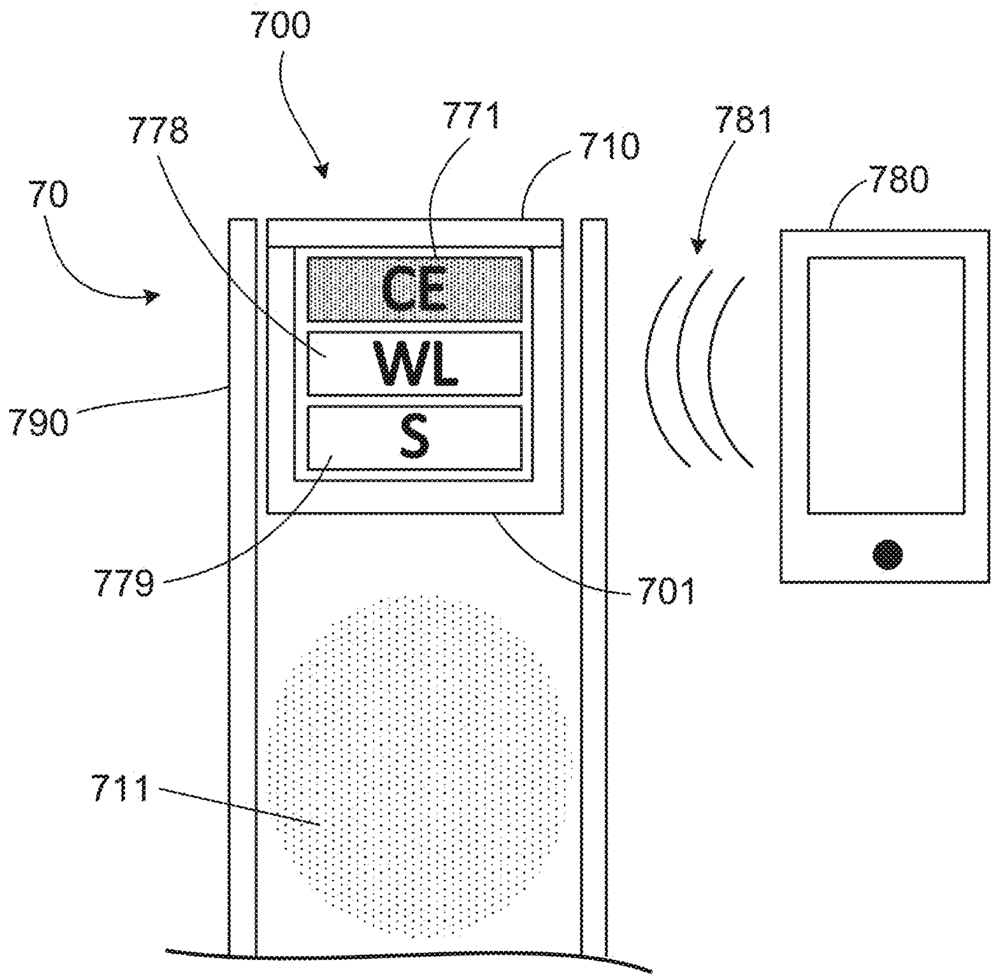
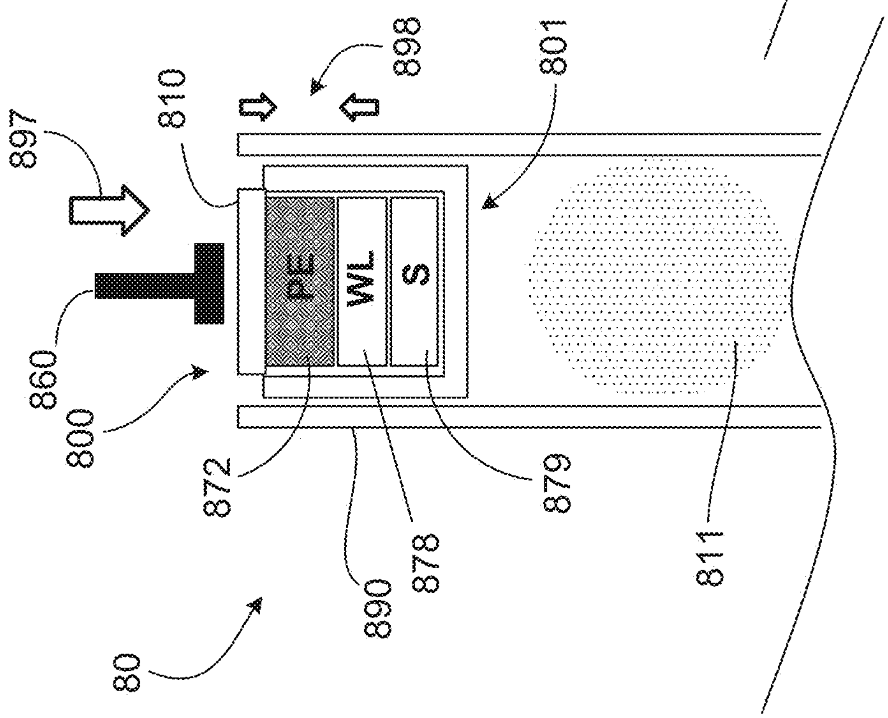


FIG. 7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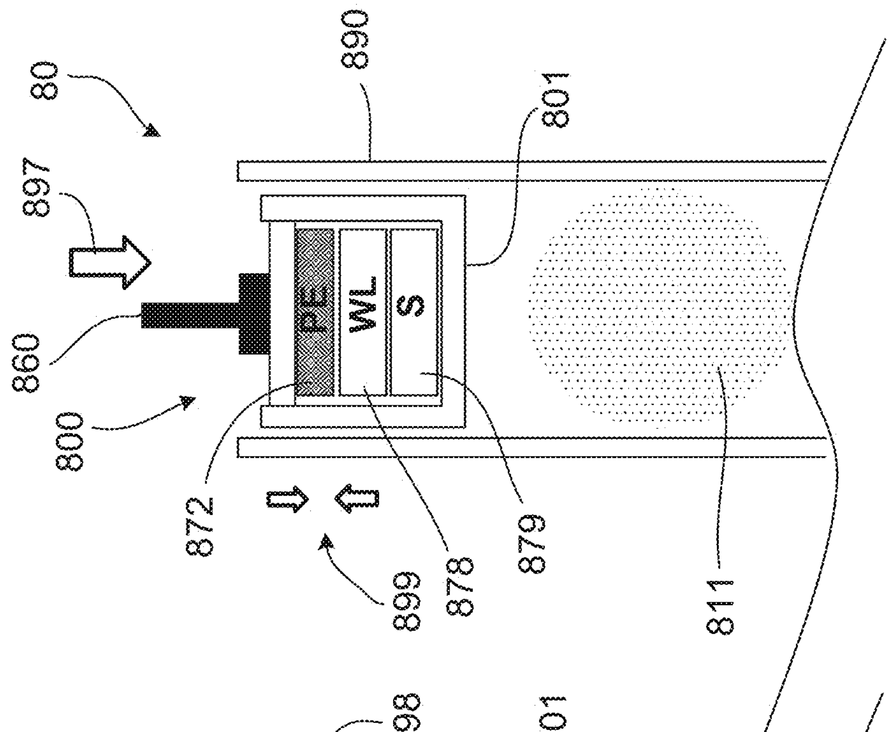
811

FIG. 8A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801
PE
WL
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872
899
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879
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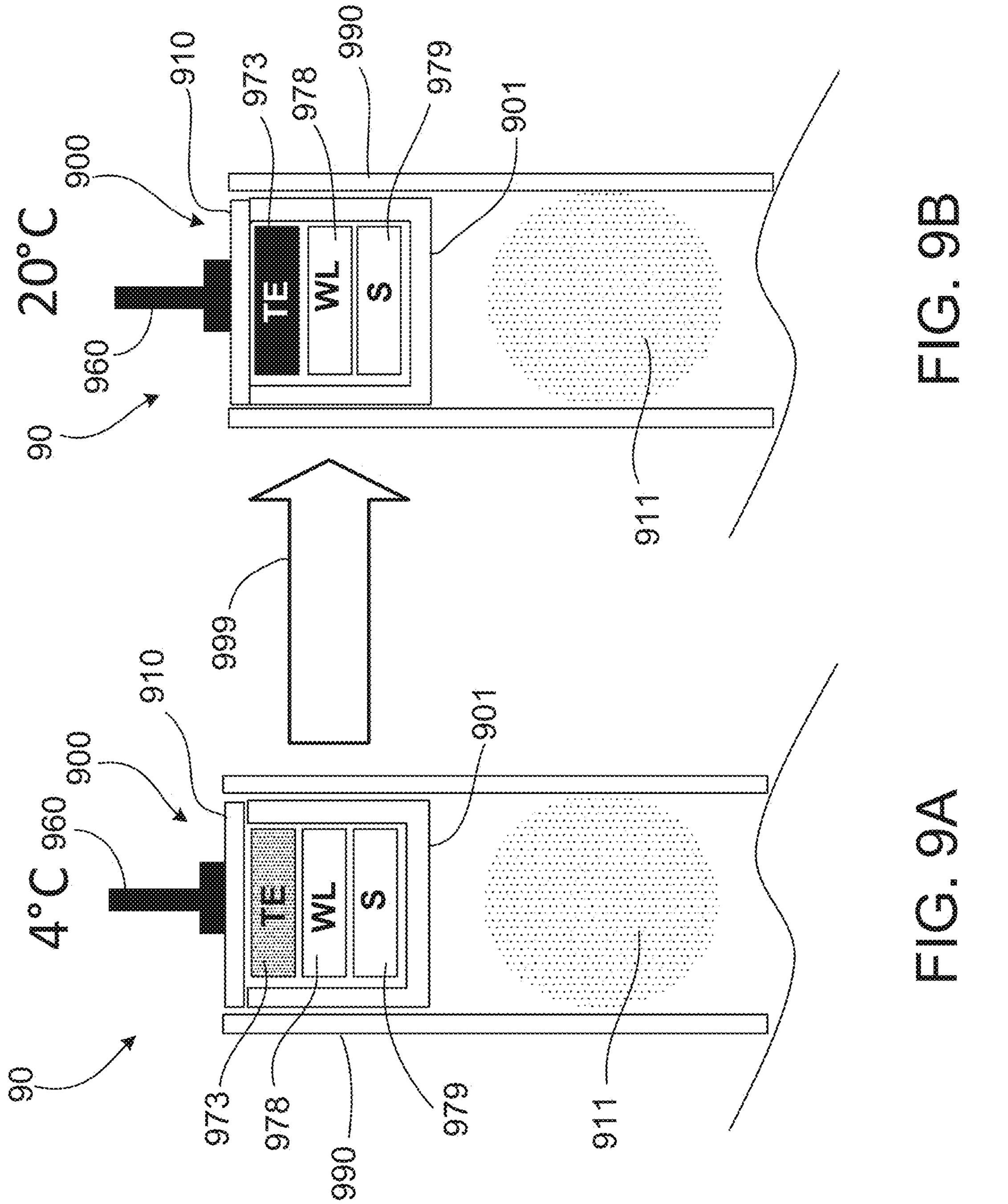
FIG. 8B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910
90
TE
WL
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901
973
978
990
979
911
999
20°C
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973
978
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979
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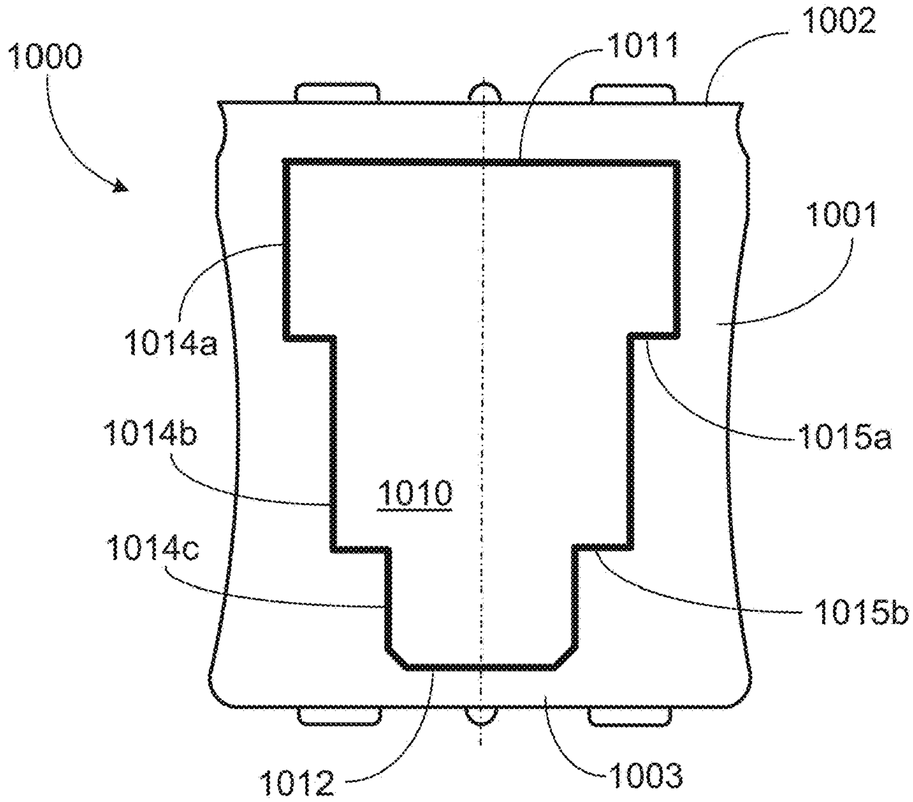
911

FIG. 9A

FIG. 9B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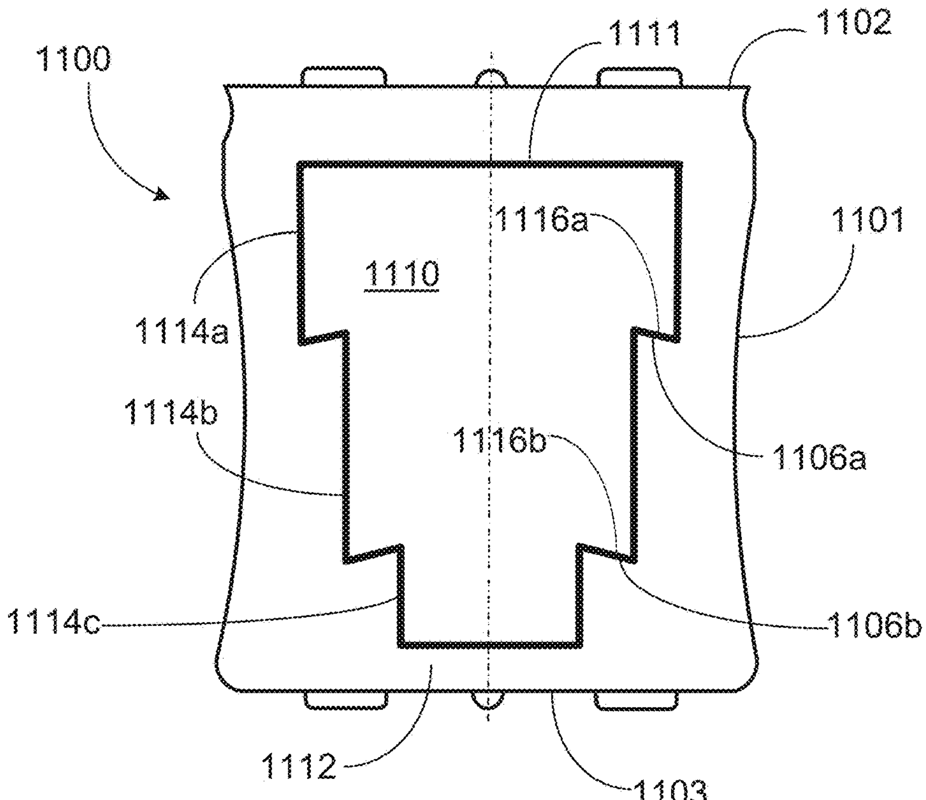
1010
1014c
1015b
1012
1003

FIG. 10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1106a
1114c
1106b
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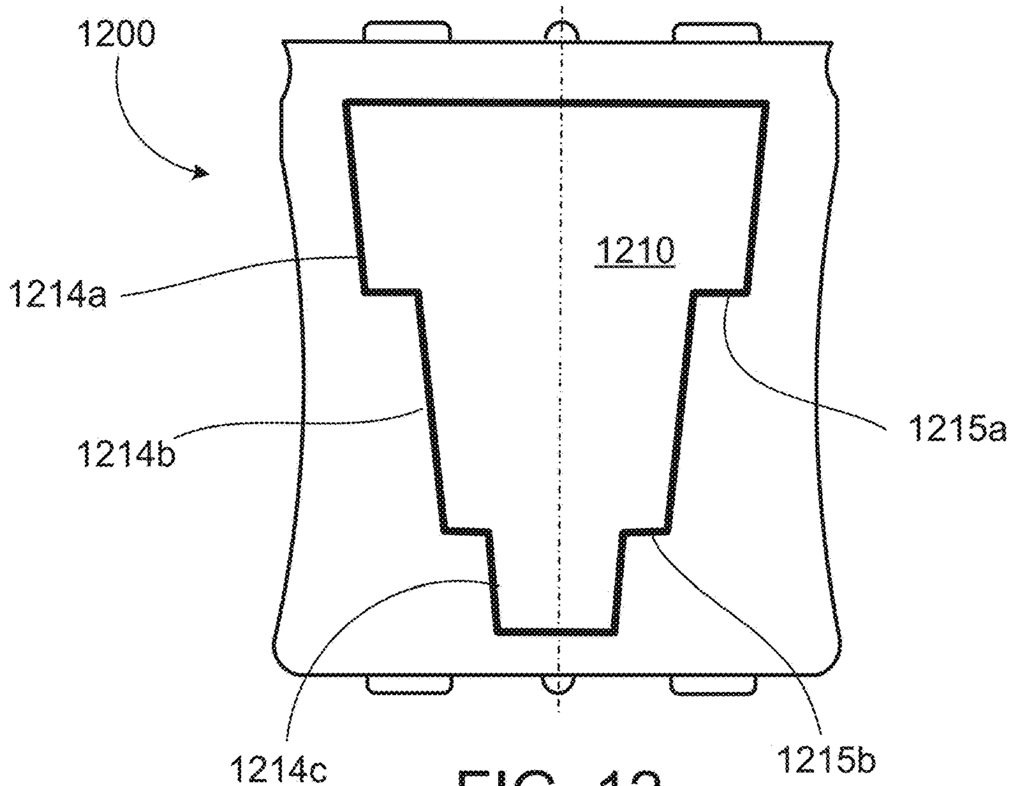
1103

FIG. 11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1210
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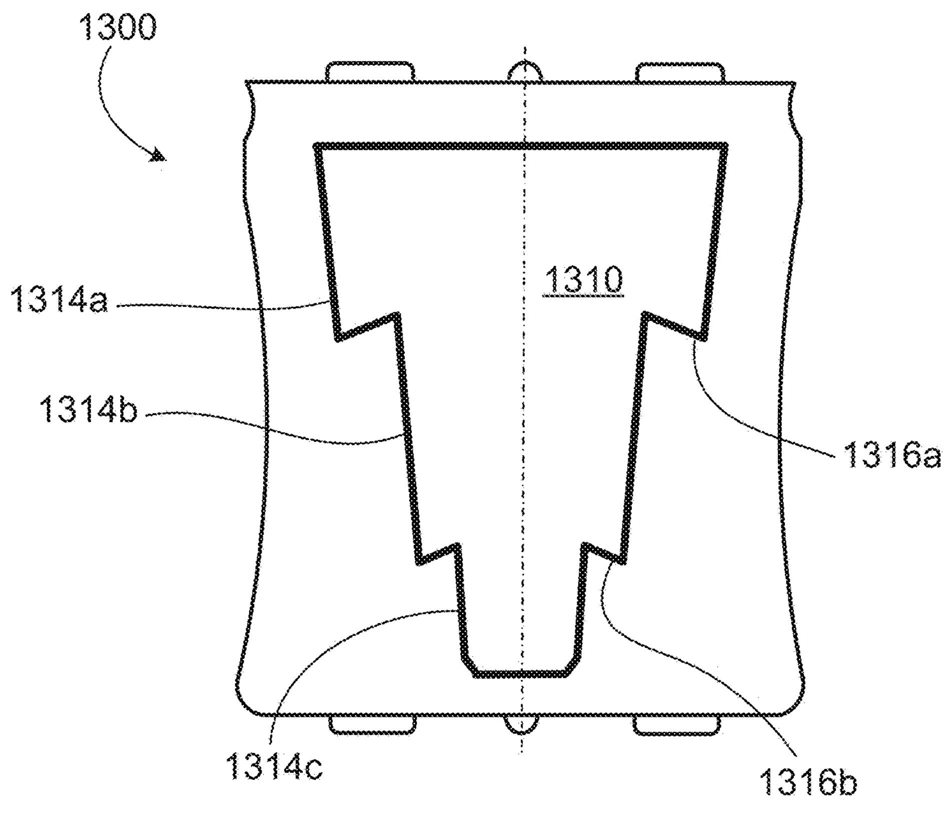
1214c
1215b

FIG. 12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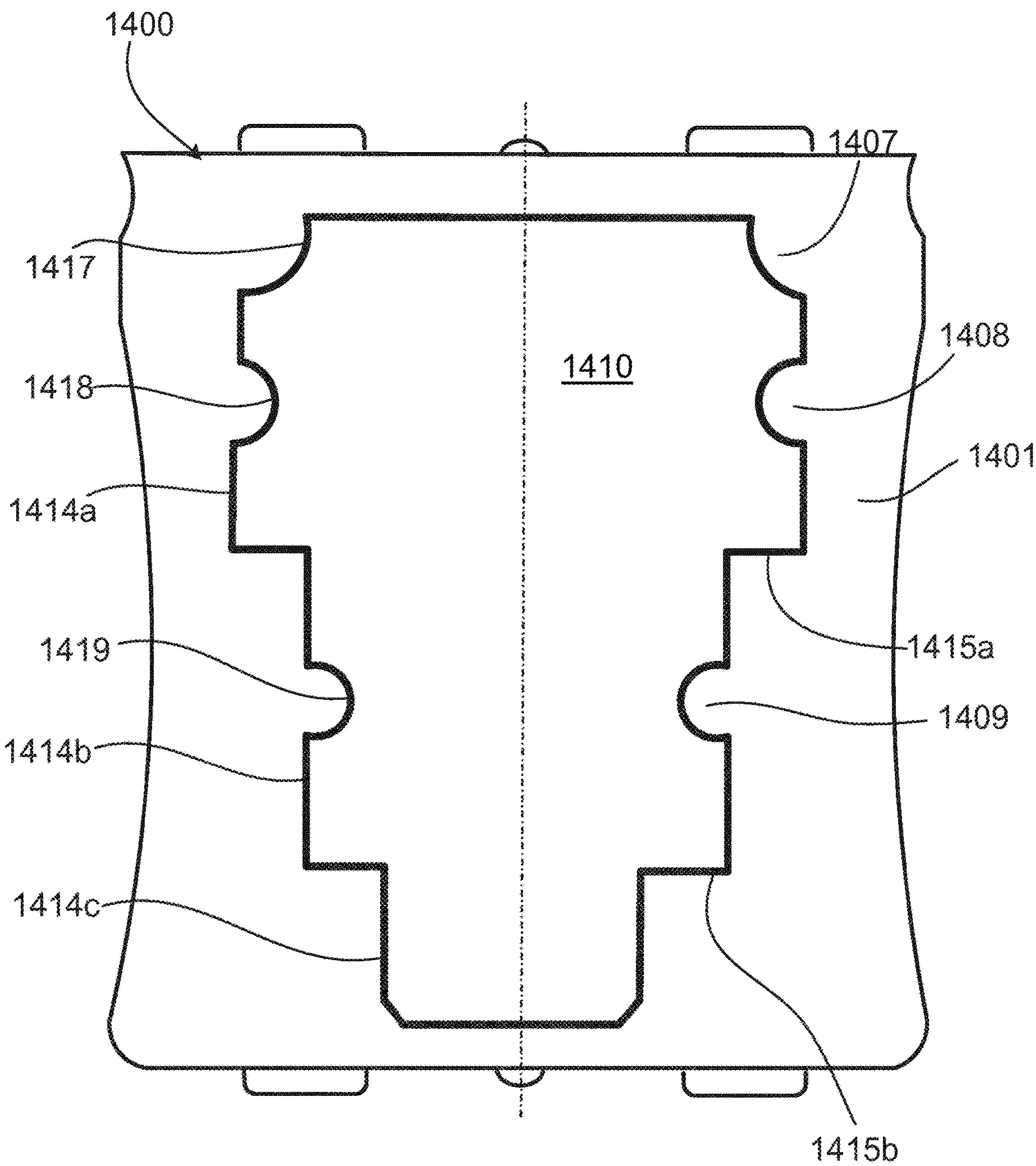
1314c
1316b

FIG. 13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1414c
1415b

FIG. 14

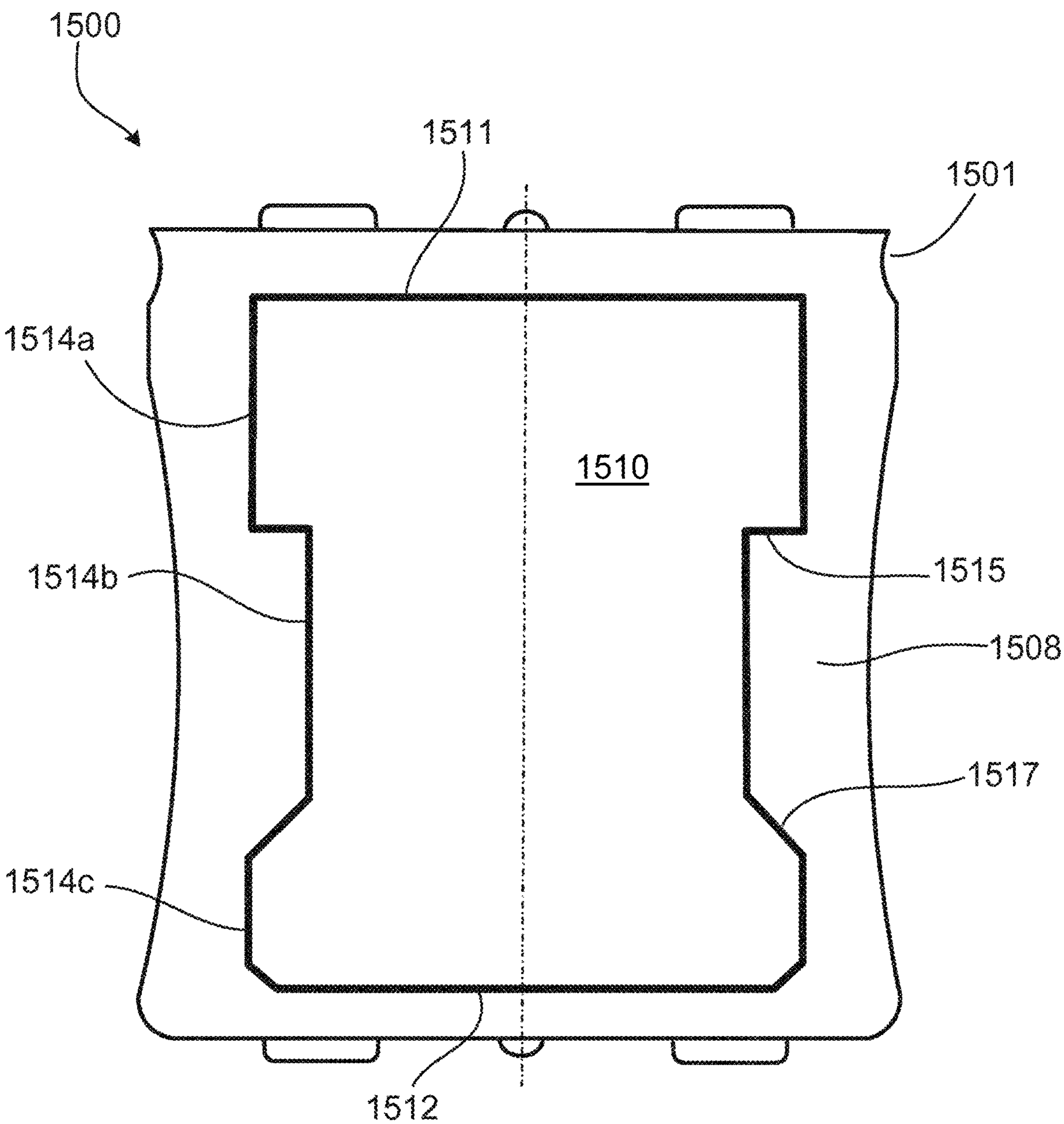

FIG. 15

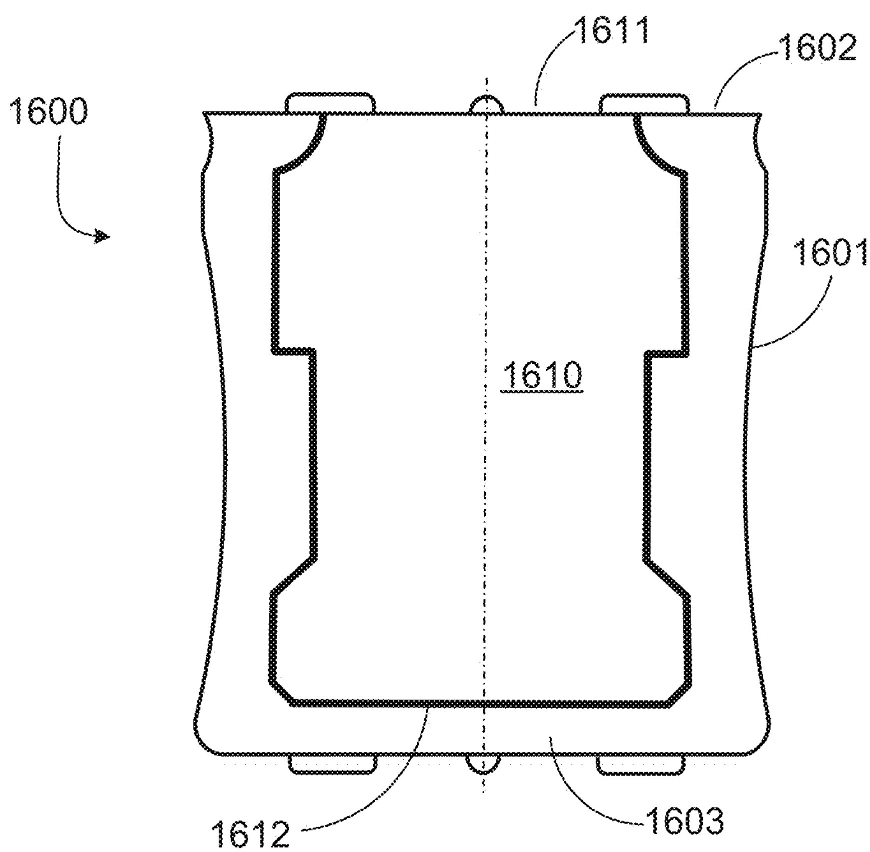

FIG. 16A

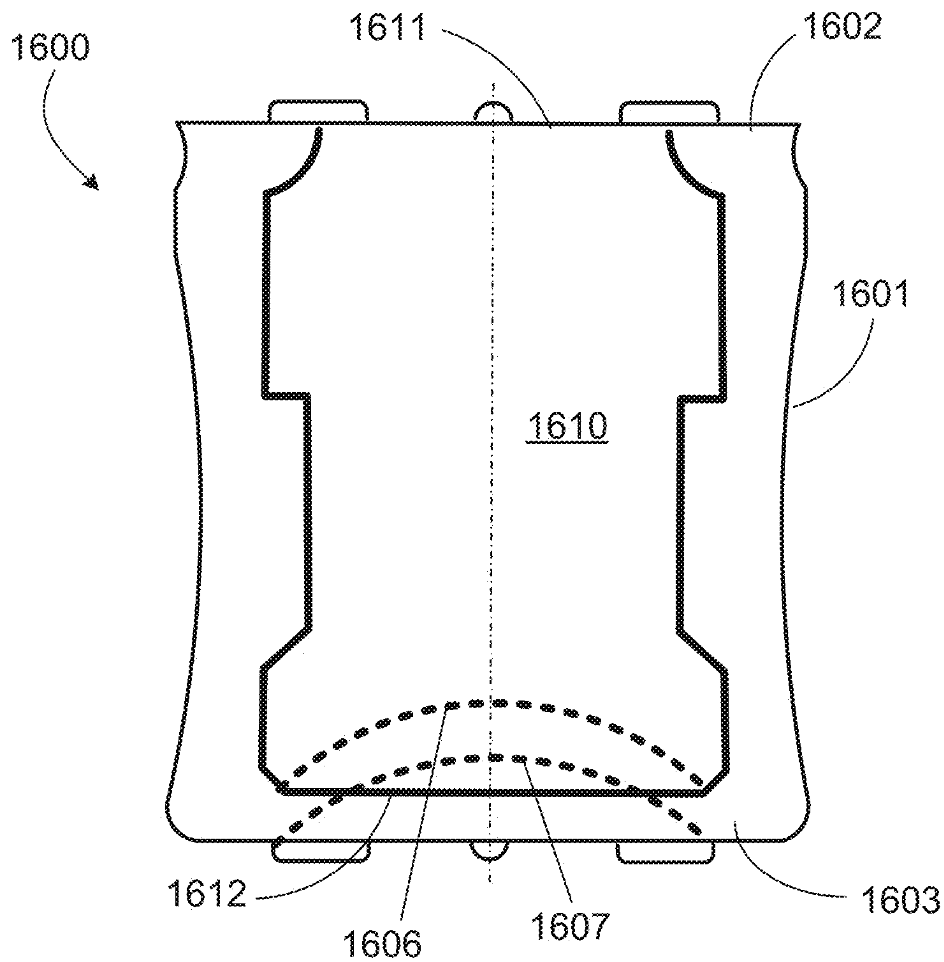

FIG. 16B

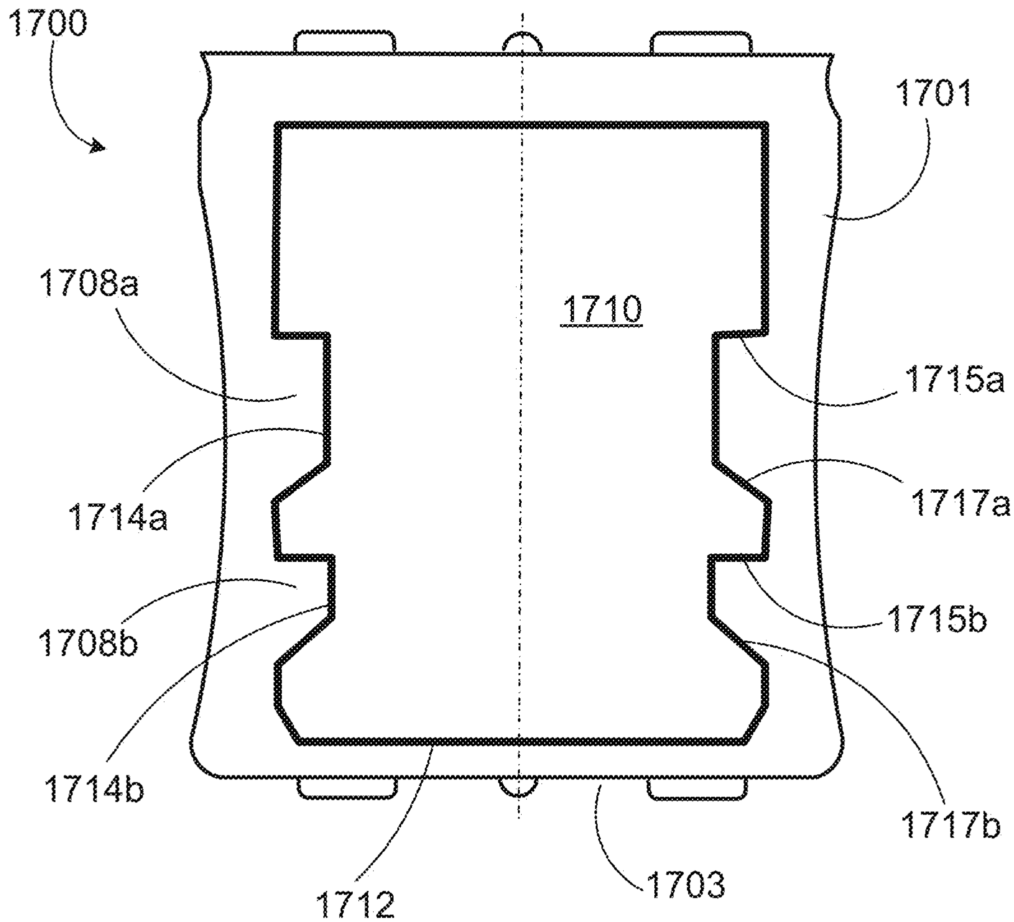
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FIG. 17

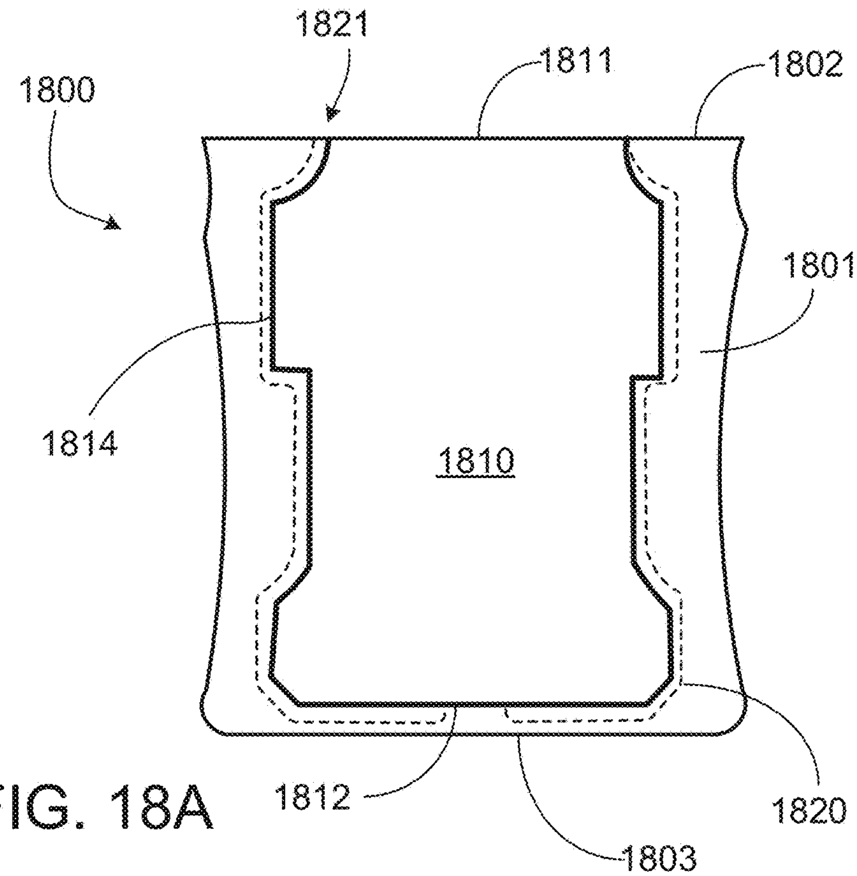

FIG. 18A

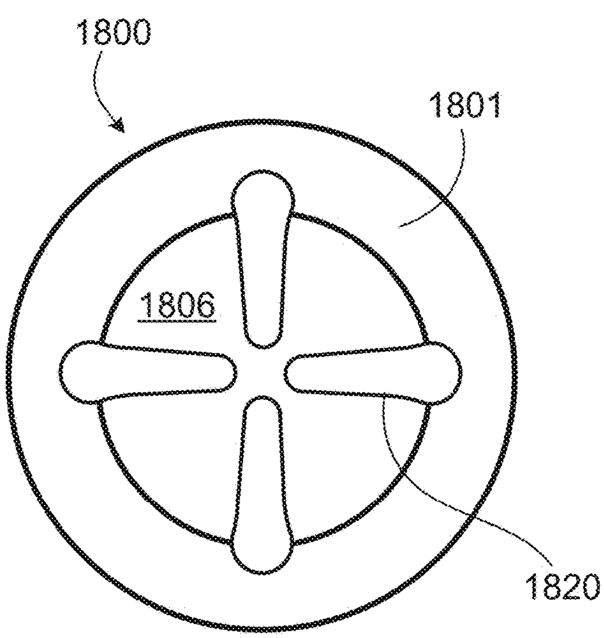

FIG. 18B

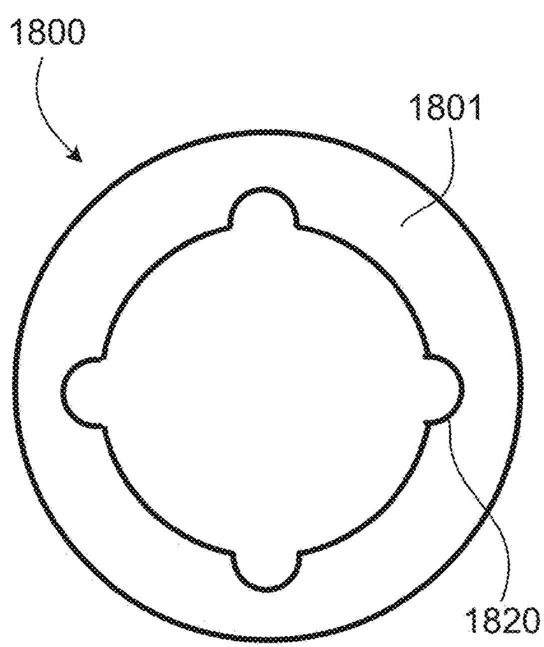

FIG. 18C

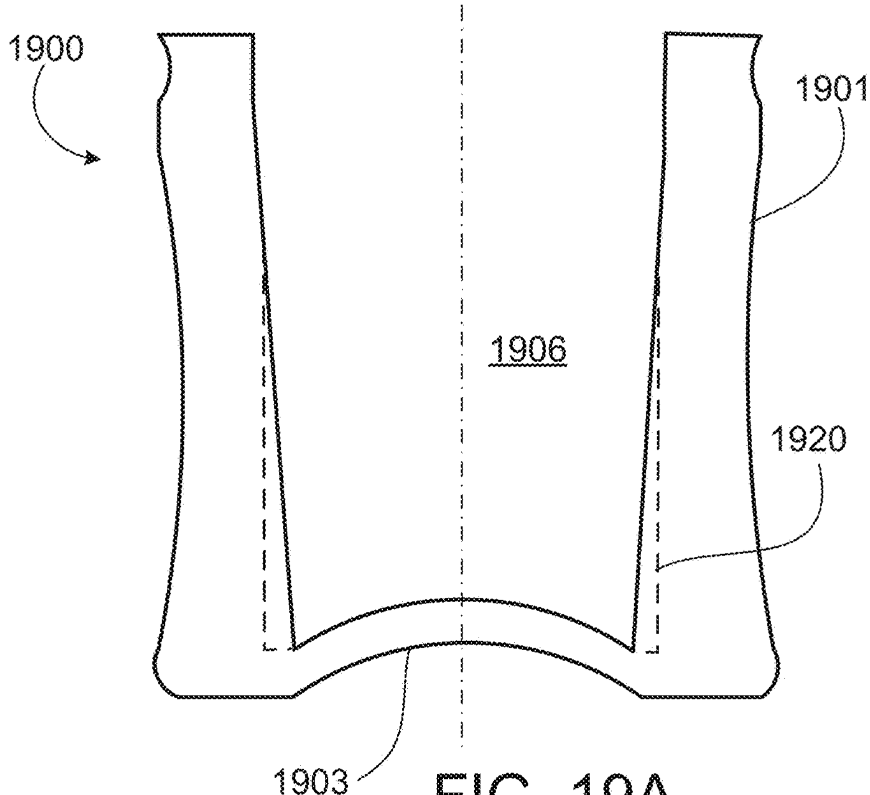

FIG. 19A

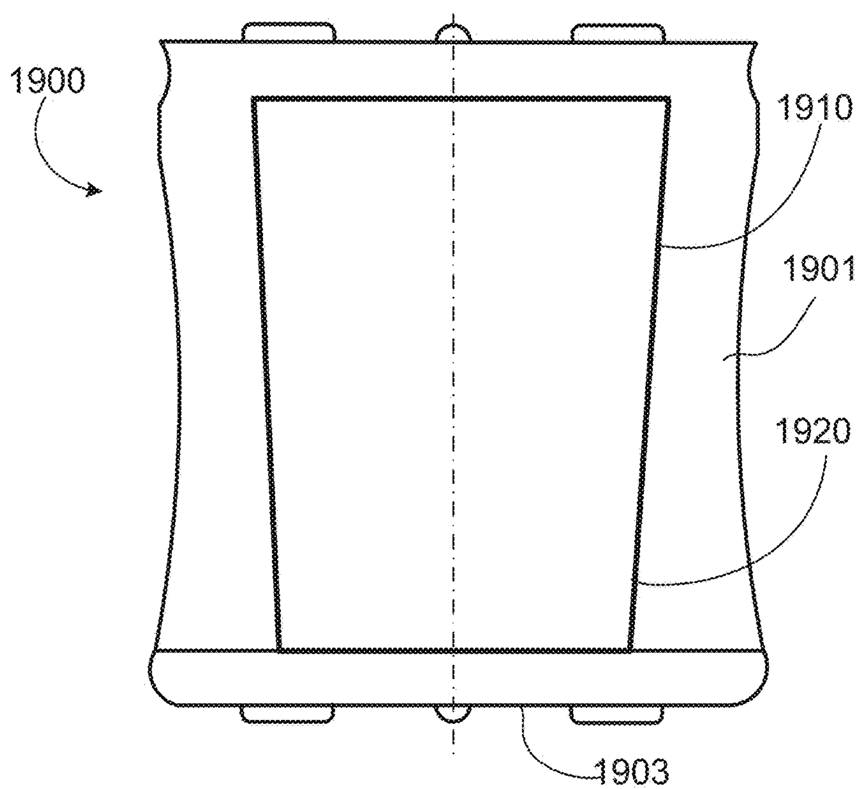

FIG. 19B

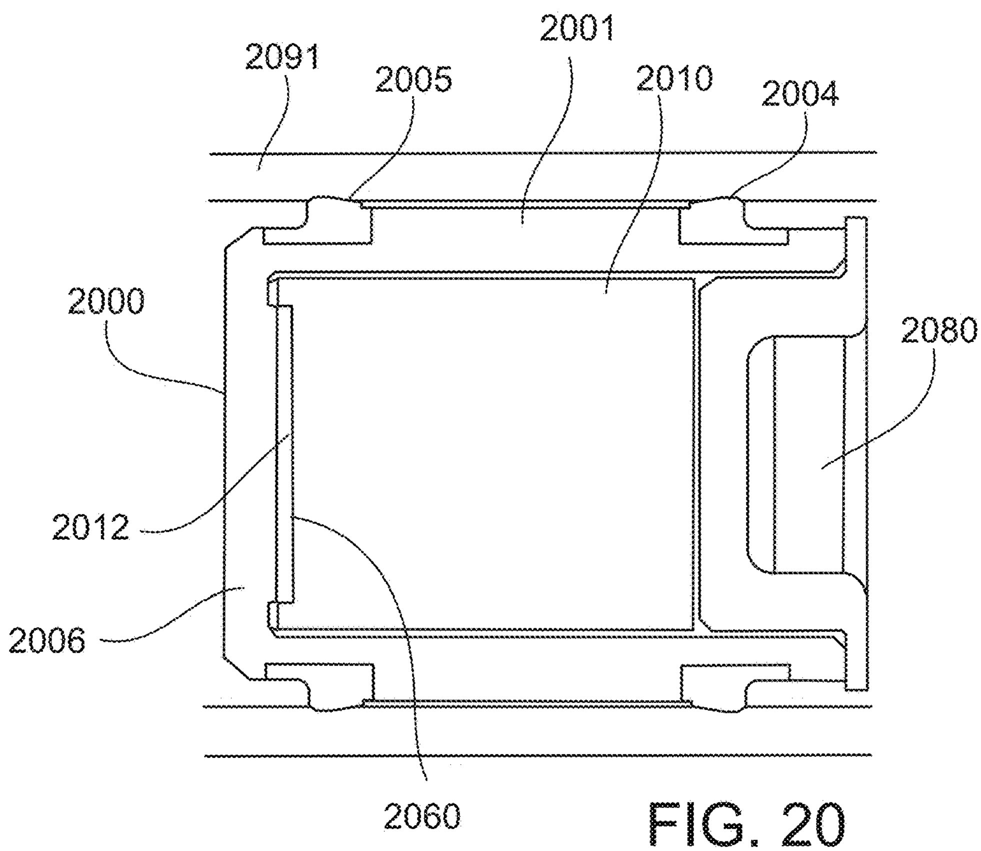
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FIG. 20

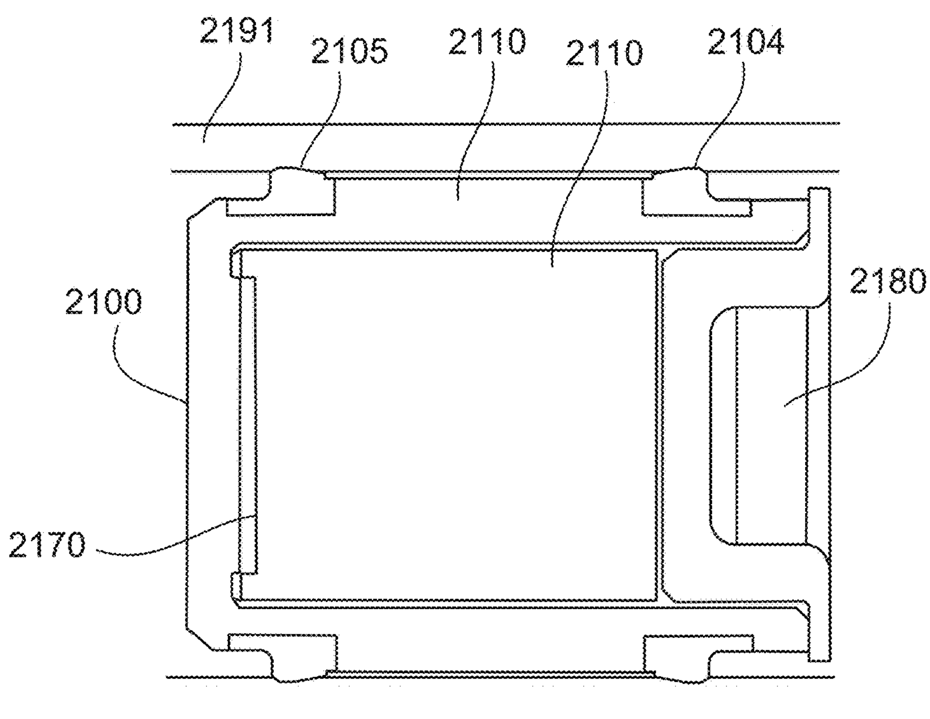
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FIG. 21

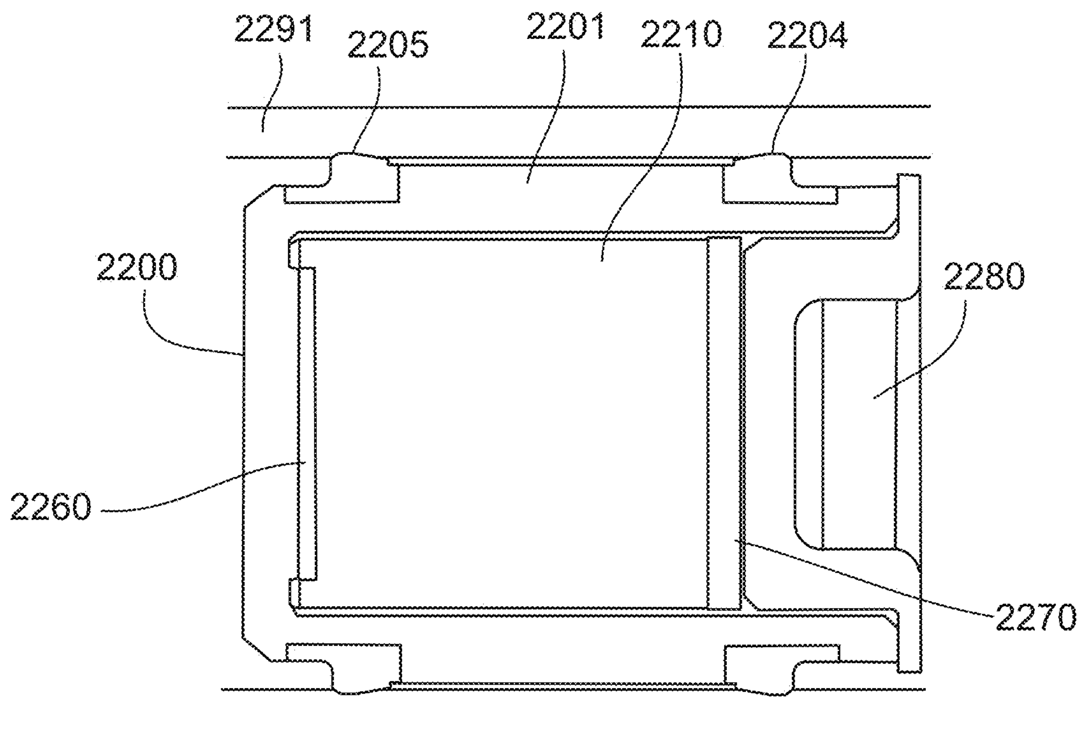

FIG. 22

STOPPER

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. application Ser. No. 16/968,735, filed on Aug. 10, 2020, which is the national stage entry of International Patent Application No. PCT/EP2019/053195, filed on Feb. 8, 2019, and claims priority to Application No. EP 18305141.6, filed on Feb. 12, 2018, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] This description relates to a stopper for use in a cartridge or syringe of a medical device configured to eject a medicament using the stopper.

BACKGROUND

[0003] A variety of diseases exist that require treatment by injection of a medicament. Such injection can be performed using injection devices, which are applied either by medical personnel or by patients themselves. As an example, type-1 and type-2 diabetes can be treated by patients themselves by injection of insulin doses, for example once or several times per day. For instance, a pre-filled disposable insulin pen or autoinjector can be used as an injection device. Alternatively, a re-usable pen or autoinjector may be used. A re-usable pen or autoinjector allows replacement of an empty medicament cartridge by a new one. Each of these devices typically employs an elastomeric stopper or bung to drive a medicament from the cartridge or a syringe in the device and some include one or more electronic devices embedded in the stopper.

SUMMARY

[0004] An example disclosure of the present embodiment is a stopper for use in a cartridge or syringe of a medical device, the stopper configured to be disposed within a container closure system, the stopper comprising a shell comprising a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, the open end defining a cavity and the sidewalls defining an exterior surface sized and shaped to fit inside the container closure system; and an insert configured to be inserted into the cavity, the insert sized and shaped to receive a force from a plunger rod and distribute the force to the shell in order to advance the shell into the container closure system, wherein the a closed end of the shell in the cavity defines a convex surface configured to be contacted and deflected by the insert upon insertion into the cavity, and wherein the closed end of the shell is made of an elastic and/or plastic deformable material.

[0005] The closed end of the shell may define an arch region defining the convex surface in the cavity and a concave surface of the exterior of the closed end of the shell, and the arch region may be configured to be deflected by the insert contacting the convex surface during insertion into the cavity.

[0006] The cavity may define inwardly tapering sidewalls from the open end to the closed end of the shell, and the inwardly tapering sidewalls may be configured to maintain an inward taper when the arch region is deflected by the insert.

[0007] The cavity may comprise an interior step element, and the insert may comprise a corresponding step element configured to abut the interior step element of the shell and distribute at least a portion of the force from the plunger rod to the interior step element.

[0008] The cavity may define a snap-fit feature extending into the cavity, the snap-fit feature configured to retain the insert in the cavity by being deflected or deformed by the insert during insertion of the insert into the cavity until the snap-fit feature relaxes into a corresponding depression in the insert, and an exterior surface of the insert may define the corresponding depression being sized and positioned to accept the snap-fit feature.

[0009] The corresponding step element may be sized and positioned to retain the interior step element and resist radial deflection of the interior step element by the force applied to the insert.

[0010] The interior step element may define a first interface surface, and the corresponding step element may define a second interface surface, and insertion of the insert into the shell may abut the first interface surface against the second interface surface.

[0011] The first interface surface and the second interface surface may define acute angles about the closed end of the shell.

[0012] The cavity may comprise at least one venting channel extending from the closed end of the shell towards the open end of the shell, the at least one venting channel being sized and positioned to enable air and/or fluid in the cavity to be expelled though the at least one venting channel during insertion of the insert into the cavity.

[0013] The at least one venting channel may extend partially towards the open end of the shell and not extend fully to the open end of the shell.

[0014] The insert may define a conically tapering exterior surface configured to enable venting of air in the cavity around the exterior of the insert during insertion of the insert.

[0015] The insert may comprise an electronic device having a sensor configured to generate a sensing signal, and the closed end of the shell may be configured to pass the sensing signal therethrough.

[0016] The sensor may be configured to be responsive to position of the stopper in the container closure system.

[0017] Another example embodiment of the present disclosure is a container closure system comprising a cartridge or syringe housing and a stopper as described above, wherein the shell is configured to be inserted into the housing prior to the container closure system being filled with the medical product, and the insert is configured to be inserted into the stopper after the filling procedure.

[0018] The container closure system may comprise a syringe, and the cartridge housing may be a housing of the syringe, and the insert may be disposed at a distal end of plunger rod of the syringe configured to be inserted into the shell of the stopper after the syringe is assembled into a medical device.

[0019] The housing may be configured for use with one or more of: an autoinjector, a pen injector, and an injection pump.

[0020] The cartridge or syringe housing may contain a liquid medicament.

[0021] Another example embodiment of the present disclosure is a stopper configured to be disposed within a container closure system. The stopper includes a shell and an insert configured to be inserted into a cavity of the shell. The shell includes a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, the open end defining a cavity and the sidewalls defining an exterior surface sized and shaped to fit inside the container closure system, with the cavity having an interior step element. The insert is sized and shaped to receive a force from a plunger rod and distribute the force to the shell in order to advance the shell into the container closure system. The insert includes a corresponding step element configured to abut the interior step element of the shell and distribute at least a portion of the force from the plunger rod to the interior step element.

[0022] In some examples, the cavity defines a snap-fit feature extending into the cavity, the snap-fit feature configured to retain the insert in the cavity by being deflected or deformed by the insert during insertion of the insert into the cavity until the snap-fit feature relaxes into a corresponding depression in the insert, and wherein an exterior surface of the insert defines the corresponding depression being sized and positioned to accept the snap-fit feature.

[0023] In some examples, the snap-fit feature extends radially around the cavity.

[0024] In some examples, the corresponding depression extends radially around the exterior surface of the insert.

[0025] In some examples, the corresponding step element is sized and positioned to retain the interior step element and resist radial deflection of the interior step element by the force applied to the insert.

[0026] In some examples, the interior step element defines a first interface surface, and wherein the corresponding step element defines a second interface surface, and wherein insertion of the insert into the shell abuts the first interface surface against the second interface surface. In some examples, the first and second interface surfaces define an acute angle about the closed end of the shell.

[0027] In some examples, the insert is sized and shaped to fit completely within the cavity when inserted.

[0028] In some examples, the insert, when inserted, defines a contact surface above the open end of the shell with respect to the longitudinal axis.

[0029] In some examples, the cavity defines a venting channel extending from the open end of the cavity to a rearward end of the cavity opposite the open end, the venting channel being sized and positioned to enable air and/or fluid in the cavity to be expelled though the venting channel during insertion of the insert into the cavity. In some examples, the venting channel extends radially along at least a length of the rearward end of the cavity.

[0030] In some examples, the insert defines a conically tapering exterior surface configured to enable venting of air in the cavity around the exterior of the insert during insertion of the insert.

[0031] In some examples, the insert includes an electronic device having a sensor configured to generate a sensing signal, wherein the closed end of the shell is configured to pass the sensing signal therethrough.

[0032] In some examples, the sensor is configured to be responsive to position of the stopper in the container closure system.

[0033] In some examples, in the insert includes a cap member configured to seal the open end of the cavity against the insert when the insert is inserted into the cavity.

[0034] In some examples, the insert is configured to be secured to the shell using one or more of the following: a snap-fit feature, glue, welding, ultrasonic welding, friction welding, or thermal welding.

[0035] In some examples, the shell is constructed from a material that can be sterilized.

[0036] In some examples, the insert is a distal end of the plunger rod.

[0037] In some examples, the stopper includes a sealing member disposed around an exterior surface of the sidewalls and arranged to form a seal between the exterior surface and an inner surface of the container closure system when the stopper is disposed within the container closure system.

[0038] In some examples, the shell is a soft shell including a substantially pliable material.

[0039] In some examples, an exterior surface of the sidewalls of the soft shell defines a sealing region radially around the exterior surface and arranged to form a seal between the exterior surface and an inner surface of the container closure system when the stopper is disposed within the container closure system.

[0040] In some examples, wherein the closed end of the shell defines a convex shape in the cavity configured to be contacted and deflected by the insert upon insertion into the cavity.

[0041] In some examples, the closed end of the shell defines an arch region defining the convex shape in the cavity and a concave surface of the exterior of the closed end of the shell, and wherein the arch region is configured to be deflected by the insert upon insertion into the cavity.

[0042] Another example is a container closure system including a cartridge or syringe housing and a stopper configured to be disposed within the container closure system. The stopper includes a shell and an insert configured to be inserted into a cavity of the shell. The shell includes a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, the open end defining a cavity and the sidewalls defining an exterior surface sized and shaped to fit inside the container closure system, with the cavity having an interior step element. The insert is sized and shaped to receive a force from a plunger rod and distribute the force to the shell in order to advance the shell into the container closure system. The insert includes a corresponding step element configured to abut the interior step element of the shell and distribute at least a portion of the force from the plunger rod to the interior step element. The shell is configured to be inserted into the cartridge or syringe prior to the container closure system is being filled with the medical product, and the insert is configured to be inserted into the stopper after the filling procedure.

[0043] In some examples, the medical cartridge is a syringe, and the cartridge housing is a housing of the syringe, and wherein the insert is a disposed at the distal end of the plunger rod of the syringe configured to be inserted into the shell of the stopper after the syringe is assembled into a medical device.

[0044] In some examples, the housing is configured for use with an autoinjector.

[0045] In some examples, the housing is configured for use with a pen injector

[0046] In some examples, the housing is configured for use with an injection pump.

[0047] In some instances, the medicament includes a pharmaceutically active compound.

[0048] Another example is a stopper configured to be disposed within a container closure system. The stopper includes a shell having a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell and an insert configured to be inserted into a cavity of the stopper, the insert being sized and shaped to receive a force from a plunger rod and distribute the force to the shell in order to advance the shell into the container closure system. The open end of the stopper defines the cavity and the sidewalls define an exterior surface sized and shaped to fit inside the container closure system. The closed end of the shell defines an arch region defining a convex surface at a closed end of the cavity and a concave surface of the exterior of the closed end of the shell, and wherein the arch region is configured to be deflected by the insert contacting the convex region upon insertion into the cavity.

[0049] In some instances, the cavity defines inwardly tapering sidewalls from the open end to the closed end of the cavity, and wherein the inwardly tapering sidewalls are configured to maintain an inward taper when the arch region is deflected by the insert.

[0050] Yet another example is a stopper configured to be disposed within a container closure system, where the stopper includes a shell having a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, where the open end defines a cavity and the sidewalls define an exterior surface sized and shaped to fit inside an inner surface of the container closure system. The stopper includes a sealing element arranged around the exterior surface of the side walls, an insert configured to be inserted into the cavity, and a closure cap configured to be inserted into the cavity behind the insert and seal the insert inside the cavity, where the sealing element is configured to create a seal between the shell and the inner surface of the container closure system.

[0051] In some instances, the stopper includes an adhesive element configured to be disposed in the cavity at a closed end of the cavity and to adhere the insert to the shell, the adhesive element configured to receive a distal end of the insert during insertion of the insert into the cavity or during insertion of the closure cap into the cavity against a proximal end of the insert.

[0052] In some instances, the stopper includes a deformable element configured to be disposed in the cavity between the insert and the closure cap, where the deformable element is configured to be deformed by a distal end of the closure cap during insertion of the closure cap into the cavity.

[0053] In some instances, the stopper includes a deformable element configured to be disposed in the cavity at a closed end of the cavity, where the deformable element is configured to be deformed by a distal end of the insert during insertion of the insert into the cavity or during insertion of the closure cap into the cavity against a proximal end of the insert.

[0054] In general, the examples described herein relate to pharmaceutical closure components with a sensor insert for positioning an embedded electronics assembly into an operative location inside a cartridge or syringe cylinder.

BRIEF DESCRIPTION OF THE FIGURES

[0055] FIG. **1** is an exploded view of an injection device.

[0056] FIG. **2**A is a side view of a stopper configured to be disposed in a cartridge of an injection device.

[0057] FIG. **2**B is a cross sectional schematic of the stopper of FIG. **2**A disposed in a cartridge.

[0058] FIG. **3** is a cross-sectional view of a stopper with a stepped insert disposed in a cavity of the stopper.

[0059] FIG. **4** is cross-sectional view of a stopper with an insert secured in a cavity of the stopper with snap-fit features.

[0060] FIG. **5** is cross-sectional view of a stopper with an insert disposed in a cavity of the stopper.

[0061] FIG. **6**A is cross-sectional view of a stepped insert being inserted into a corresponding stepped cavity of a stopper.

[0062] FIG. **6**B is cross-sectional view the assembled stopper of FIG. **6**A containing the insert before being driven by a plunger.

[0063] FIG. **6**C is cross-sectional view the stopper of FIG. **6**A during a sensing operating during or after being driven into the cartridge by the plunger

[0064] FIG. **7** is cross-sectional view of a stopper disposed within a cartridge and powered by a wireless system.

[0065] FIGS. **8**A and **8**B are cross-sectional views of a stopper disposed within a cartridge and powered by a piezoelectric system.

[0066] FIGS. **9**A and **9**B are cross-sectional views of a stopper disposed within a cartridge and powered by a thermoelectric system.

[0067] FIG. **10** is a side view of a stopper showing the outline of an insert disposed in the stopper, where the insert has a stepped profile.

[0068] FIG. **11** is a side view of a stopper showing the outline of an insert disposed in the stopper, where the insert has a stepped profile with acute angles.

[0069] FIG. **12** is a side view of a stopper showing the outline of a tapered insert disposed in the stopper, where the insert has a stepped profile.

[0070] FIG. **13** is a side view of a stopper showing the outline of a tapered insert disposed in the stopper, where the insert has a stepped profile with acute angles.

[0071] FIG. **14** is a side view of a stopper showing the outline of a insert disposed in the stopper, where the insert has a stepped profile with snap-fit features.

[0072] FIG. **15** is a side view of a stopper showing the outline of an insert disposed in the stopper, where the insert is fully inserted into the stopper.

[0073] FIG. **16**A is a side view of a stopper showing the outline of an insert disposed in the stopper, where the insert is flush with the stopper.

[0074] FIG. **16**B is a side view of the stopper of FIG. **16** prior to insertion of the insert where the front part of the rubber has a concave shape towards the inner cavity.

[0075] FIG. **17** is a side view of a stopper showing the outline of an insert disposed in the stopper, where the insert is sized to contain a larger electronics assembly.

[0076] FIG. **18**A is a cross-sectional view of a stopper with an insert deposed in a cavity of the stopper, where the cavity includes a vent channel.

[0077] FIG. **18**B is a top view of the stopper of FIG. **18**A with the insert removed to view the vent channels in the cavity.

[0078] FIG. 18C is a cross-sectional view of the stopper of FIG. 18A showing vent channels though a middle portion of the shell.

[0079] FIG. 19A is a cross-sectional view of a stopper with a conical cavity and an arched distal end.

[0080] FIG. 19B is a side view of the stopper of FIG. 19A showing the outline of an insert disposed in the stopper, where the arched distal end is at a deflected position.

[0081] FIG. 20 is a cross-section illustration of rigid shell with in-molded seals where an insert is secured inside a cavity of the rigid shell with an adhesive.

[0082] FIG. 21 is a cross-section illustration of rigid shell with in-molded seals where an insert is pressed into a deformable material in a cavity of the rigid shell.

[0083] FIG. 22 is cross-section illustration of a rigid shell with in-molded seals where an insert secured inside a cavity of the rigid shell with an adhesive and a deformable material.

DETAILED DESCRIPTION

[0084] Information on the use of medical devices by patients can be determined by sensors that generate data, which can then be accessed through wireless communication. For injection devices, one or more sensors may be integrated into the stopper of a drug containing cartridge or syringe (also referred to as a plunger or plunger stopper). Wireless communication of the data requires an energy source, e.g., a battery cell, that also needs to be connected to the sensor and thus may be integrated into the stopper. Typical manufacturing processes for sterile plunger stoppers to be used in injection devices include process steps with high temperatures (e.g., rubber molding and steam sterilization processes), which may not be compatible with some electronic components (e.g., batteries). Therefore, some cartridge-based injection and medical syringe systems can be difficult to sterilize prior to use if they include electronics in the replaceable portions of the device (e.g., a cartridge stopper or a cartridge housing). In some instances, elements containing electronic assemblies may be separated from the plunger stopper to be assembled at a later stage after the sterilization processes are completed.

[0085] Electronic devices and assemblies may contain temperature sensitive materials (e.g., polymers) which may get degrade after heat is applied. Further, batteries can lose performance at high temperatures. To avoid exposure of the electronics components to excess heat, in some examples, part of the cartridge stopper (sometimes referred to as a bung) can be sterilized before the electronics are added. For example, a stopper shell can be forming a seal against a cartridge wall, and an insert containing electronic devices separate from the stopper can be used so that the insert is assembled into the stopper after a heat sterilization step (and, therefore, also without impacting the sterilized liquid inside the cartridge). In some instances, the insert is integrated into the distal end of a plunger rod of an injection device and is inserted into a stopper shell prior to use of the injection device. In some examples, electronic components (e.g., RFID sensors) can be added to a previously heat-sterilized stopper of a disposable or reusable drug cartridge.

[0086] Stoppers used in a drug containing cartridge or syringe typically fulfill functional requirements such as achieving dosing precision, maintaining container closure integrity, and providing certain force profiles with a feedback to the patient. Container closure systems, generally, include a cartridge or a syringe composed of a barrel, typically made of glass or plastic and a stopper plus a closure cap. As described in detail below, a shell of the stopper may define a cavity on the open or proximal end (the end opposite the drug-contacting end) configured to accept a sensor insert (also referred to simply as an insert) that, in some instances, contains functional electronic devices. In some instances, the insert being disposed in the cavity places an electronic device in close proximity to the volume enclosed by the stopper and thereby enables a sensor in the insert to conduct sensing operations of the volume (e.g.,., determining the location of the stopper in the cartridge). Assembling a sensor insert into a stopper may lead to the following issues that influence the functional requirements:

[0087] (i) Elastic deformation of the stopper under the impact of an injection force may affect dosing precision,

[0088] (ii) Movement of the sensor insert from its resting position in the stopper may affect dosing precision,

[0089] (iii) Misalignment of the sensor insert may tilt the stopper and affect container closure integrity,

[0090] (iv) Misalignment of the sensor insert may tilt the stopper and affect force profiles during an injection,

[0091] (v) Misalignment of the sensor insert may affect an ultrasonic sound emission and reception from a sensing unit in the insert, and thus jeopardize precise position measurement of the stopper in a cartridge or syringe,

[0092] (vi) Air inclusions between the sensor insert and the stopper may induce oscillation during a dosing operation and subsequently affect dosing precision, and

[0093] (vii) Air inclusion in front of a sensor insert may affect functionality of e.g. ultrasound measurement principles and thus leading to wrong results.

[0094] Accordingly, techniques described herein include stoppers having a shell with a cavity configured to accept a sensor insert to achieve appropriate injection force transmission, acceptable alignment and fixation of components, and avoid air inclusion, among other things. In operation, an injection force applied by a plunger rod presses onto the back end of a stopper and pushes the stopper forward in a cartridge, thus expelling a medicament from the cartridge. The cavity is formed in the back end of the stopper shell and a sensor insert is disposed in the cavity. With an insert in the shell, the plunger rod transmits the force to the sensor insert and, in some instances, the insert has a stepped shape that defines interface surfaces between each step, such that the force of the plunger on the insert is distributed to the stopper in a staggered manner through the interface surfaces against corresponding surfaces in the cavity of the stopper. In some embodiments, the stepped profile of the insert includes features configured to interface with the cavity and radially retain the cavity against the insert. In some instances, these features are configured to strengthen the radial fixation of the elastomer stopper to the insert during the transmission of the force from the insert to the stopper, and, as a result, reduce radial deflection of the cavity (i.e., away from the insert).

[0095] In some instances, once the sensor insert is assembled in its determined position in the stopper, it is fixed in-place with snap-fit elements. Such snap-fit elements may be circumferential rings of rubber material that match with corresponding groves in the sensor insert element. One snap-fit element at the back end of the stopper and optionally further snap-fit elements can be applied.

[0096] In some instances, a sensor insert includes a circular shape with a smaller diameter at a distal section that is oriented towards the distal end of the stopper (i.e., the drug-contacting end), and a larger diameter of the proximal section that is at the proximal end of the stopper. In some instances, the distal section of the sensor insert contains a sensor element and the proximal section contains a power source (e.g., a battery cell) connected to the sensor element. Additional electronic components (e.g. an integrated circuit or a printed circuit board) may be placed in between (or integrated with) the sensor element and the power source, or in a different configuration, which is, in some instances, determined by the preferred location of the sensor element in the stopper in order to conduct a sensing operation. In some instances, the cavity inside the stopper narrows down from the distal to the proximal end and provides guidance for the insertion of the sensor insert. In one embodiment, the sensor insert is cone-shaped with a matching design of the stopper inner cavity to allow insertion of the sensor insert without friction between rubber material of the stopper and the sensor insert until the snap-fit features engage to retain the sensor insert in place in the cavity. In an alternate embodiment, threads may be used in addition to snap-fit features to lock the sensor insert into position.

[0097] In addition to the cone-shape design of the interface, in some instances, one or more lateral channels to allow air to vent before becoming trapped in dead ends are provided. In addition, a front end of the cavity may have a convex shape extending into the cavity to allow the insert to make a first central contact with the rubber shell to expel entrapped air from the dead end. In some instances, the closed end of the stopper defines an arch shape that includes the convex shape of the end of the cavity a concave exterior surface of the distal end of the shell. In some instances, the closed end of the cavity includes extending lateral channels configured to vent the air from the cavity when the insert is inserted.

[0098] The terms "drug" or "medicament" are used herein to describe one or more pharmaceutically active compounds. As described below, a drug or medicament can include at least one small or large molecule, or combinations thereof, in various types of formulations, for the treatment of one or more diseases. Exemplary pharmaceutically active compounds may include small molecules; polypeptides, peptides and proteins (e.g., hormones, growth factors, antibodies, antibody fragments, and enzymes); carbohydrates and polysaccharides; and nucleic acids, double or single stranded DNA (including naked and cDNA), RNA, antisense nucleic acids such as antisense DNA and RNA, small interfering RNA (siRNA), ribozymes, genes, and oligonucleotides. Nucleic acids may be incorporated into molecular delivery systems such as vectors, plasmids, or liposomes. Mixtures of one or more of these drugs are also contemplated.

[0099] The terms "drug" or "medicament" are used synonymously herein and describe a pharmaceutical formulation containing one or more active pharmaceutical ingredients or pharmaceutically acceptable salts or solvates thereof, and optionally a pharmaceutically acceptable carrier. An active pharmaceutical ingredient ("API"), in the broadest terms, is a chemical structure that has a biological effect on humans or animals. In pharmacology, a drug or medicament is used in the treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being. A drug or medicament may be used for a limited duration, or on a regular basis for chronic disorders.

[0100] As described below, a drug or medicament can include at least one API, or combinations thereof, in various types of formulations, for the treatment of one or more diseases. Examples of API may include small molecules having a molecular weight of **500** Da or less; polypeptides, peptides and proteins (e.g., hormones, growth factors, antibodies, antibody fragments, and enzymes); carbohydrates and polysaccharides; and nucleic acids, double or single stranded DNA (including naked and cDNA), RNA, antisense nucleic acids such as antisense DNA and RNA, small interfering RNA (siRNA), ribozymes, genes, and oligonucleotides. Nucleic acids may be incorporated into molecular delivery systems such as vectors, plasmids, or liposomes. Mixtures of one or more drugs are also contemplated.

[0101] The drug or medicament may be contained in a primary package or "drug container" adapted for use with a drug delivery device. The drug container may be, e.g., a cartridge, syringe, reservoir, or other solid or flexible vessel configured to provide a suitable chamber for storage (e.g., short-or long-term storage) of one or more drugs. For example, in some instances, the chamber may be designed to store a drug for at least one day (e.g., 1 to at least 30 days). In some instances, the chamber may be designed to store a drug for about 1 month to about 2 years. Storage may occur at room temperature (e.g., about 20° C.), or refrigerated temperatures (e.g., from about −4° C. to about 4° C.). In some instances, the drug container may be or may include a dual-chamber cartridge configured to store two or more components of the pharmaceutical formulation to-be-administered (e.g., an API and a diluent, or two different drugs) separately, one in each chamber. In such instances, the two chambers of the dual-chamber cartridge may be configured to allow mixing between the two or more components prior to and/or during dispensing into the human or animal body. For example, the two chambers may be configured such that they are in fluid communication with each other (e.g., by way of a conduit between the two chambers) and allow mixing of the two components when desired by a user prior to dispensing. Alternatively or in addition, the two chambers may be configured to allow mixing as the components are being dispensed into the human or animal body.

[0102] The drugs or medicaments contained in the drug delivery devices as described herein can be used for the treatment and/or prophylaxis of many different types of medical disorders. Examples of disorders include, e.g., diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy, thromboembolism disorders such as deep vein or pulmonary thromboembolism. Further examples of disorders are acute coronary syndrome (ACS), angina, myocardial infarction, cancer, macular degeneration, inflammation, hay fever, atherosclerosis and/or rheumatoid arthritis. Examples of APIs and drugs are those as described in handbooks such as Rote Liste 2014, for example, without limitation, main groups 12 (anti-diabetic drugs) or 86 (oncology drugs), and Merck Index, 15th edition.

[0103] Examples of APIs for the treatment and/or prophylaxis of type 1 or type 2 diabetes mellitus or complications associated with type 1 or type 2 diabetes mellitus include an insulin, e.g., human insulin, or a human insulin analogue or derivative, a glucagon-like peptide (GLP-1), GLP-1 analogues or GLP-1 receptor agonists, or an analogue or derivative thereof, a dipeptidyl peptidase-4 (DPP4) inhibitor, or a pharmaceutically acceptable salt or solvate thereof, or any mixture thereof. As used herein, the terms “analogue” and “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, by deleting and/or exchanging at least one amino acid residue occurring in the naturally occurring peptide and/or by adding at least one amino acid residue. The added and/or exchanged amino acid residue can either be codable amino acid residues or other naturally occurring residues or purely synthetic amino acid residues. Insulin analogues are also referred to as “insulin receptor ligands”. In particular, the term „derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, in which one or more organic substituent (e.g. a fatty acid) is bound to one or more of the amino acids. Optionally, one or more amino acids occurring in the naturally occurring peptide may have been deleted and/or replaced by other amino acids, including non-codeable amino acids, or amino acids, including non-codeable, have been added to the naturally occurring peptide.

[0104] Examples of insulin analogues are Gly(A21), Arg (B31), Arg(B32) human insulin (insulin glargine); Lys(B3), Glu(B29) human insulin (insulin glulisine); Lys(B28), Pro (B29) human insulin (insulin lispro); Asp(B28) human insulin (insulin aspart); human insulin, wherein proline in position B28 is replaced by Asp, Lys, Leu, Val or Ala and wherein in position B29 Lys may be replaced by Pro; Ala(B26) human insulin; Des(B28-B30) human insulin; Des(B27) human insulin and Des(B30) human insulin.

[0105] Examples of insulin derivatives are, for example, B29-N-myristoyl-des(B30) human insulin, Lys(B29) (N-tetradecanoyl)-des(B30) human insulin (insulin detemir, Levemir®); B29-N-palmitoyl-des(B30) human insulin; B29-N-myristoyl human insulin; B29-N-palmitoyl human insulin; B28-N-myristoyl LysB28ProB29 human insulin; B28-N-palmitoyl-LysB28ProB29 human insulin; B30-N-myristoyl-ThrB29LysB30 human insulin; B30-N-palmitoyl-ThrB29LysB30 human insulin; B29-N-(N-palmitoyl-gamma-glutamyl)-de(B30) human insulin, B29-N-omega-carboxypentadecanoyl-gamma-L-glutamyl-des(B30) human insulin (insulin degludec, Tresiba®); B29-N-(N-lithocholyl-gamma-glutamyl)-des(B30) human insulin; B29-N-(ω-carboxyheptadecanoyl)-des(B30) human insulin and B29-N-(ω-carboxyheptadecanoyl) human insulin.

[0106] Examples of GLP-1, GLP-1 analogues and GLP-1 receptor agonists are, for example, Lixisenatide (Lyxumia®), Exenatide (Exendin-4, Byetta®, Bydureon®, a 39 amino acid peptide which is produced by the salivary glands of the Gila monster), Liraglutide (Victoza®), Semaglutide, Taspoglutide, Albiglutide (Syncria®), Dulaglutide (Trulicity®), rExendin-4, CJC-1134-PC, PB-1023, TTP-054, Langlenatide/HM-11260C, CM-3, GLP-1 Eligen, ORMD-0901, NN-9924, NN-9926, NN-9927, Nodexen, Viador-GLP-1, CVX-096, ZYOG-1, ZYD-1, GSK-2374697, DA-3091 March-701, MAR709, ZP-2929, ZP-3022, TT-401, BHM-034. MOD-6030, CAM-2036, DA-15864, ARI-2651, ARI-2255, Exenatide-XTEN and Glucagon-Xten.

[0107] An examples of an oligonucleotide is, for example: mipomersen sodium (Kynamro®), a cholesterol-reducing antisense therapeutic for the treatment of familial hypercholesterolemia.

[0108] Examples of DPP4 inhibitors are Vildagliptin, Sitagliptin, Denagliptin, Saxagliptin, Berberine.

[0109] Examples of hormones include hypophysis hormones or hypothalamus hormones or regulatory active peptides and their antagonists, such as Gonadotropine (Follitropin, Lutropin, Choriongonadotropin, Menotropin), Somatropine (Somatropin), Desmopressin, Terlipressin, Gonadorelin, Triptorelin, Leuprorelin, Buserelin, Nafarelin, and Goserelin.

[0110] Examples of polysaccharides include a glucosaminoglycane, a hyaluronic acid, a heparin, a low molecular weight heparin or an ultra-low molecular weight heparin or a derivative thereof, or a sulphated polysaccharide, e.g. a poly-sulphated form of the above-mentioned polysaccharides, and/or a pharmaceutically acceptable salt thereof. An example of a pharmaceutically acceptable salt of a poly-sulphated low molecular weight heparin is enoxaparin sodium. An example of a hyaluronic acid derivative is Hylan G-F 20 (Synvisc®), a sodium hyaluronate.

[0111] The term “antibody”, as used herein, refers to an immunoglobulin molecule or an antigen-binding portion thereof. Examples of antigen-binding portions of immunoglobulin molecules include F(ab) and F(ab')2 fragments, which retain the ability to bind antigen. The antibody can be polyclonal, monoclonal, recombinant, chimeric, de-immunized or humanized, fully human, non-human, (e.g., murine), or single chain antibody. In some embodiments, the antibody has effector function and can fix complement. In some embodiments, the antibody has reduced or no ability to bind an Fc receptor. For example, the antibody can be an isotype or subtype, an antibody fragment or mutant, which does not support binding to an Fc receptor, e.g., it has a mutagenized or deleted Fc receptor binding region. The term antibody also includes an antigen-binding molecule based on tetravalent bispecific tandem immunoglobulins (TBTI) and/or a dual variable region antibody-like binding protein having cross-over binding region orientation (CODV).

[0112] The terms “fragment” or “antibody fragment” refer to a polypeptide derived from an antibody polypeptide molecule (e.g., an antibody heavy and/or light chain polypeptide) that does not comprise a full-length antibody polypeptide, but that still comprises at least a portion of a full-length antibody polypeptide that is capable of binding to an antigen. Antibody fragments can comprise a cleaved portion of a full length antibody polypeptide, although the term is not limited to such cleaved fragments. Antibody fragments that are useful in the present disclosure include, for example, Fab fragments, F(ab')2 fragments, scFv (single-chain Fv) fragments, linear antibodies, monospecific or multispecific antibody fragments such as bispecific, trispecific, tetraspecific and multispecific antibodies (e.g., diabodies, triabodies, tetrabodies), monovalent or multivalent antibody fragments such as bivalent, trivalent, tetravalent and multivalent antibodies, minibodies, chelating recombinant antibodies, tribodies or bibodies, intrabodies, nanobodies, small modular immunopharmaceuticals (SMIP), binding-domain immunoglobulin fusion proteins, camelized antibodies, and VHH containing antibodies. Additional examples of antigen-binding antibody fragments are known in the art.

[0113] The terms "Complementarity-determining region" or "CDR" refer to short polypeptide sequences within the variable region of both heavy and light chain polypeptides that are primarily responsible for mediating specific antigen recognition. The term "framework region" refers to amino acid sequences within the variable region of both heavy and light chain polypeptides that are not CDR sequences, and are primarily responsible for maintaining correct positioning of the CDR sequences to permit antigen binding. Although the framework regions themselves typically do not directly participate in antigen binding, as is known in the art, certain residues within the framework regions of certain antibodies can directly participate in antigen binding or can affect the ability of one or more amino acids in CDRs to interact with antigen.

[0114] Examples of antibodies are anti PCSK-9 mAb (e.g., Alirocumab), anti IL-6 mAb (e.g., Sarilumab), and anti IL-4 mAb (e.g., Dupilumab).

[0115] Pharmaceutically acceptable salts of any API described herein are also contemplated for use in a drug or medicament in a drug delivery device. Pharmaceutically acceptable salts are for example acid addition salts and basic salts.

[0116] FIG. **1** is an exploded view of an injection device **100**, which may be a disposable or reusable injection device. The injection device **100** includes a housing **110** and a having therein a cartridge **114** with cartridge housing **104**, to which a needle assembly **115** can be affixed. The needle **109** of needle assembly **115** is protected by an inner needle cap **116** and an outer needle cap **117**, which in turn can be covered by cap **118**. A medicament or drug dose to be ejected from the injection device **100** is selected by turning a dosage knob **112**, and the selected dose is displayed via a dosage window or display **113**. The display can include a digital display, for example. The dosage display **113** relates to the section of the injection device through or on which the selected dosage is visible.

[0117] As described further below, the injection device **100** may include one or more electronic components **122**, **124**, some of which may be included in the stopper **108**, for example.

[0118] Turning the dosage knob **112** causes a mechanical click sound to provide acoustical feedback to a user. The numbers displayed in the dosage display **113** are printed on a sleeve that is contained in the housing **110** and mechanically interacts with a stopper in the cartridge **114**. When the needle **109** is stuck into a skin portion of a patient, and then the injection button **111** is pushed, the drug dose displayed in the display **113** will be ejected from the injection device **100**. During an injection, a drive mechanism **106**, which is shown as an outline of a plunger arm, drives a stopper **108** into the cartridge to expel the drug. The stopper is an important element of the container closure system because it is a barrier for fluids leaking in and out; a gas barrier (e.g., oxygen), and prevents evaporation of H2O and other fluids. In some embodiments, the seal function is provided by sealing elements (**604**, **605** of FIG. **6**A) in contact with the container walls, but still allows the stopper to glide. When the needle **109** of injection device **100** remains for a certain time in the skin portion after the injection button **111** is pushed, a high percentage of the dose is actually injected into the patient's body.

[0119] FIG. **2**A is a side view of a stopper **200** configured to be disposed in a cartridge of an injection device. The stopper **200** includes a proximal end **202** configured to be contacted by a plunger rod or other drive device in, for example, a syringe, injector, auto-injector, infusion pump device, or similar. The stopper also includes a distal end **203**, which is a drug-contacting end of the stopper **200** and configured to be in contact with a medicament or similar inside of a cartridge or syringe into which the stopper **200** is disposed. The stopper **200** includes a body **201**, which is later referred to as a shell when it defines an interior cavity configured to accept an insert, and an exterior surface of the body **201** includes proximal and distal sealing element **204**,**205** formed into the exterior surface as regions of a larger diameter, which are configured to sealingly interface with an inner surface of the cartridge of syringe into which the stopper **200** is disposed in order to seal the medicament into the cartridge or syringe and maintain that seal during movement of the stopper **200** along the inner surface, as illustrated in FIGS. **2**B and **6**B below. The sealing interface may form at least part of a sterile barrier within the cartridge **290** (FIG. **2**B), which is required to preserve the sterility of the medicament to be delivered by an injection device.

[0120] FIG. **2**B is a cross sectional view of a container closure system including the stopper **200** of FIG. **2**A disposed in a cartridge **290**. As shown in FIG. **2**B, the stopper **200** is placed in the cartridge **290** such that the proximal and distal sealing element **204**, **205** formed into the exterior surface of the body **201** of the stopper are in sealing engagement with an inner surface of a housing **291** of the cartridge **290**. The cartridge **290** and stopper **200** enclose a volume **299** in the cartridge **290**, and a distal end of the cartridge is closed by a cap **292**, which may be, in some instances, a septum or port configured to deliver a medicament contained in the volume **299** when the stopper **200** is advanced in the cartridge **290**.

[0121] FIG. **3** is a cross-sectional view of a stopper **300** illustrating a stepped insert **310** disposed in a cavity of a shell **301** of the stopper **300**. The shell **301** includes a distal end **303** and a proximal end **302**, with the cavity (not visible) formed in the proximal end **302**, where the insert **310** is shown occupying the volume of the cavity. The insert includes a proximal end **311** being flush with the proximal end of the shell **301**, and a distal end **312** located close to the distal end of the shell **301**. The insert **310** is constructed of three main part **313***a-c*, which may be segments of a unibody construction, or individual parts joined or otherwise fixed together. Each of the main parts **313***a-c* of the insert **310** defines an exterior surface **314***a-c* having a conical taper decreasing towards the distal end **312**, where discontinuities in the taper diameter between each of the exterior surfaces **314***a-c* create stepped surface **315***a*, **315***b* between the main segments **313***a-c*. The stepped surfaces **315***a*, **315***b* and the exterior surfaces **314***a-c* (and, for completeness, the distal end **312**) are mirrored by the interior surface of the cavity of the shell **301** in order to interface the insert **310** in the cavity of the shell **301** as closely as possible. In this manner, the stepped surfaces **315***a*, **315***b* (illustrated as horizontal surfaces) of the insert **310** are abutting corresponding interface surfaces of the cavity, and, when a force (e.g., from a plunger rod) is applied to the proximal end **311** of the insert **310**, a portion of the force is transferred to insert across the interface surfaces abutting the stepped surfaces **315***a*, **315***b*. In operation when a force is applied to the insert from a plunger rod to drive the stopper **300** in a cartridge, this stepped configuration reduces the radial component of the applied force that is created by the interaction of the conical exterior surfaces **314***a-c* and the cavity. Additionally, when the shell **301** is constructed from an elastomeric material, the force on the insert results in a reduced radial deflection of the shell **301** away from the insert **310** as the stepped surfaces **315***a*, **315***b* press axially (i.e., along a longitudinal axis of the stopper) against the interface surfaces of the cavity. As a result, movement of the insert **310** into the shell **301** is directly resisted by the shell **301**, which enables the distal end **312** of the insert **310** to be located closer to the distal end **303** of the shell **301** without risking puncture of the distal end **303** by the distal end of the insert when the force is applied.

[0122] FIG. **4** is cross-sectional view of a stopper **400** with an insert **410** secured in a cavity of a shell **401** of the stopper **400** with snap-fit features **408**,**409**. The shell **401** includes a distal end **403** and a proximal end **402**, with the cavity (not visible) formed in the proximal end **402**, where the insert **410** is shown occupying the volume of the cavity. The insert includes a proximal end **411** being flush with the proximal end of the shell **401**, and a distal end **412** located close to the distal end of the shell **401**. The insert **410** is constructed of three main part **413***a-c*, which may be segments of a unibody construction, or individual parts joined or otherwise fixed together. Each of the main parts **413***a-c* of the insert **310** define an exterior surface **314***a-c* of a conical taper, and the proximal main part **413***a* has formed therein two grooves **418**, **419** configured to mate with the snap-fit elements **408**, **409** (e.g., circular or elliptical ring sections protruding into the cavity of the shell **401**) and retain the insert **410** in the shell **401**. In operation, the insertion of the insert **410** in the shell **401** deflects or otherwise deforms the snap-fit elements, which, in some instances, are constructed from an elastomeric material, until the snap-fit elements **408**, **409** align with a corresponding groove **418**, **419** in the insert. When aligned, the snap-fit elements **408**, **409** relax radially inward into the grooves **418**, **419** of the insert **401**, and prevent easy removal of the insert **410** from the shell **401**.

[0123] FIG. **5** is cross-sectional view of a stopper **500** with an insert disposed in a cavity of a shell **501** of the stopper **500**, where the cavity and the insert **510** defines acute angled projections that define angled interfaces surfaces **516**, b configured radially retain the shell **501** to the insert **510**. The shell **501** includes a distal end **503** and a proximal end **502**, with the cavity (not visible) formed in the proximal end **502**, where the insert **510** is shown occupying the volume of the cavity. The insert includes a proximal end **511** being flush with the proximal end of the shell **501**, and a distal end **512** located close to the distal end of the shell **501**. The insert **510** is constructed of three main part **513***a-c*, which may be segments of a unibody construction, or individual parts joined or otherwise fixed together. Each of the main parts **513***a-c* of the insert **510** defines an exterior surface **514***a-c* of a conical taper, where discontinuities in the taper diameter between each of the exterior surfaces **514***a-c* create inwardly stepped surfaces **516***a*, **516***b* between the main segments **513***a-c*. The inwardly stepped surfaces **516***a*, **516***b* and the exterior surfaces **514***a-c* (and, for completeness, the distal end **512**) are mirrored by the interior surface of the cavity of the shell **501** in order to interface the insert **510** in the cavity of the shell **501** as closely as possible. In this manner, the inwardly stepped surfaces **516***a*, **516***b* of the insert **510** are abutting corresponding interface surfaces of the cavity, and, when a force (e.g., from a plunger rod) is applied to the proximal end **511** of the insert **510**, a portion of the force is transferred to insert across the interface surfaces abutting the inwardly stepped surfaces **516***a*, **516***b*. This configuration reduces the radial component of the applied force that is created by the interaction of the conical exterior surfaces **514***a-c* and the cavity. Further, when the shell **501** is constructed from an elastomeric material, the inwardly stepped surfaces **516***a*, **516***b* prevent radial deflection of the shell **501** away from the insert **510**. As a result, when a force is applied to the insert, movement of the insert **510** into the shell **501** is reduced compared with, for example, a contiguous conical insert, because a portion of the cavity **501** is trapped between the insert **510** and the inwardly stepped surfaces **516***a*, **516***b*, and this reduction enables the distal end **512** of the insert **510** to be located closer to the distal end **503** of the shell **501** without risking puncture of the distal end **503** by the distal end of the insert.

[0124] The materials selected for the shell **301**, **401**, **501** and the insert **310**, **410**, **510** are selected based on their hardness, elasticity, and their heat resistive or insulating properties. In some instances, the insert **310**, **410**, **510** is constructed from a material chosen to be able to be molded at a temperature below the maximum exposure temperature of any embedded electronic components, or a material able to be molded for a temperature and time and able to maintain the embedded electronic components below a minimum thermal budget of the electronic components. In some implementations, the shell **301**, **401**, **501** and the insert **310**, **410**, **510** are made of polymer materials with varying elastic properties. In some implementations, heat resistive coatings may also be applied to the insert **310**, **410**, **510** or to the shell **301**, **401**, **501** to increase heat resistance, such as, for example, a polytetrafluoroethylene (PTFE) coating. In some cases, the shell **301**, **401**, **501** is made of more rigid material which is selected to be compatible with the medicament e.g., PP, PE, COC, COP, PTFE or is made of elastomeric material, e.g. butyl rubber, halobutyl rubber, thermoplastic elastomer (TPE), silicone rubbers, polyurethane and the like at least at the distal end **303**, **403**, **503** which is in contact with the medicament.

[0125] The embedded electronic components may include, for example, a sensor, an energy source, a microcontroller, and a wireless transceiver. The sensor may be, in some instances, a sensor/transmitter device such as, for example, a piezoelectric device, an acoustic sensor, or an electromagnetic sensor. The sensor/transmitter may transmit a signal, such as, for example, an ultrasonic, acoustic, light, or other signal through the shell **301**, **401**, **501** and measure a response which may, in some instances, be used to determine the position of the stopper **200**, **300**, **400**, **500** in a cartridge **114**, **290** or if an injection of the syringe has occurred. In some instances, the response received by the sensor is provided to a controller (e.g. an embedded or an external microcontroller) which may receive the response and calculate a state of the cartridge **114**, **290**. The state of the cartridge **114**, **290** may correspond to, in some instances, a fill level of medicament in the cartridge **114**, **290** or a position of the stopper **200**, **300**, **400**, **500**. In some instances, the state of the cartridge **114**, **290** enables a measurement of an injected dose of medicament.

[0126] In some instances, the energy source is a battery, by any energy harvesting technologies which may load a capacitor or a solar source. The wireless transceiver may communicate with an external electronic device as well as with the sensor and energy source. The external electronic device, which may be the controller, may communicate data received from the sensor to an external database. The wireless transceiver may communicate using any known wireless communication technique including, for example Bluetooth, NFC, or radio frequencies.

[0127] The stoppers **300**, **400**, **500** of FIGS. **3-5** may be sterilized, for example, by using a moist heat sterilization process prior to the insertion of the insert **310**, **410**, **510**. In a typical moist heat sterilization process, a completed stopper containing the insert would be sterilized at a temperature of approximately 105 to 130 degrees Celsius for between approximately fifteen and sixty minutes. In some instances, moist heat-sterilization includes saturated steam sterilization (e.g., at >121° C./20 min). In other instances, sterilization could also be done by irradiation (e.g., gamma, or e-beam) or by gas sterilization. However, batteries typically do not withstand high temperatures above 70° C. Thus, steam and high-heat sterilization processes may not be appropriate for inserts containing batteries. Further, electronics typically are sensitive to radiation. Thus, irradiation sterilization is also not right for the electronics. Suitable sterilization processes for the combination of rubber shell with insert containing sensitive electronics or batteries include gas processes, such as Ethylene Oxide (EtO), Chlorine Dioxide (CIO2) and Nitrogen Dioxide (NO2) sterilization. These processes are typically run at temperatures below 60° C. (EtO) and below 30° C. (NO2, CIO2).

[0128] In some instances, all components (e.g., shells 301, 401, 501 and inserts 310, 410, 510) would be assembled pre-sterilized in an aseptic manufacturing area prior to filling the container. The inserts could also be inserted into the stoppers after filling of the cartridge or syringe. This requires that the stopper without insert is sufficiently mechanically stable and rigid to achieve a tight seal for a filled container. For large sensor inserts, this can be best achieved with designing the stopper shell from more rigid material such as COP, COC and the like, whereas for smaller inserts a more elastomeric material like halobutyl rubber may be preferred. If the shell is to be sterilized without an insert, then steam sterilization at >121° C./20 min is preferred. An insert that is assembled into a prefilled and stoppered cartridge does not need to be sterilized, as it does not get into contact with the drug formulation.

[0129] FIG. **6**A is cross-sectional view of a stopper **600** being assembled by inserting an insert **610** into a cavity **616** formed by the shell **601**. The cavity **616** includes an interior stepped profile that matches an exterior stepped profile of the insert **610**. The cavity **606** includes interface surfaces **607***a*, *b* configured to abut corresponding stepped surfaces **615***a*, *b*, of the insert **610**, as detailed above with respect to FIG. **3**. During assembly, the insert **610** is inserted (as illustrated by arrow **697**) into the cavity **606**. Until the stepped surfaces **615***a*, *b* of the insert **610** abut the interface surfaces **607***a*, *b*. In some instances, the stopper **600** is an elastomeric container closure. The stopper **600** may be disposed inside of a cartridge (as shown in FIG. **6**B), with proximal and distal sealing edges **604**, **605** of the stopper **600** forming a seal around to contain a medicament in the cartridge. In some instances, an electronic component **680** (or electronic assembly) is contained in the insert **610** and contains, for example: a sensor, a power source (e.g., battery), a controller, a wireless communication module (e.g., IEEE 802.15, NFC, RF, IrDA), an acoustic module, memory, an on-off switch, a thermo-sensing element, a light-sensing element, or a pressure sensor. In some instances, the electronic component **680** includes an on-off switch configured to trigger the electronic component **680** by any suitable impact on the insert **610** of the stopper (e.g., by a force from a plunger rod or by a light signal).

[0130] FIG. **6**B shows the stopper **600** before being driven by a plunger rod **660** into a cartridge **690**. The plunger rod **660** (e.g., a plunger rod and head configured to contact the stopper **600**) is, in some instances, driven by an actuator or drive mechanism of an injector (as shown in FIG. **1**) having the cartridge **690**, or is a plunger rod of a syringe where the cartridge **690** is the syringe housing. In operation, the plunger rod **660** is driven (as indicated by arrows **698**) against the insert **610** and thereby applies a force to the shell **601** to move the entire stopper **600** into the cartridge **690** in order to drive a portion of the medicament **60** from the cartridge **690**.

[0131] FIG. **6**C is a cross-sectional view the stopper **600** of FIG. **6**A during a sensing operating during or after being driven into the cartridge **690** by the plunger rod **660**. FIG. **6**C shows the plunger rod **660** contacting the insert **610** of the stopper **600** and having driven the stopper **600** into the cartridge **690** (as shown by arrows **699**). In operation, the electronic component **680** may emit a sensing signal **670**, which, in some instances, is responsive to the position of the stopper **600** in the cartridge **690** and enables the electronic component **680** to generate a signal indication of the movement of the stopper **600**.

[0132] Described below are devices and methods for providing energy to electronic circuity in cartridge systems (for example, those disclosed herein) using energy harvesting to provide an alternative to standard batteries or as a supplement to batteries.

[0133] Aspects of the systems disclosed above enable medical injectors to employ 'smart' technologies by way of an attached of embedded electronic component (e.g., RFID, sensor) to give certain features to a cartridge of an injector device (e.g., of a pen-type injector). When integrating electronics into the stopper of a cartridge, one or more components may be active (e.g., a sensor to measure certain properties of the injector or cartridge) and require an energy source, which typically could be a battery. One alternative, as described below, is to use a means of energy harvesting as a power source replacement for a battery.

[0134] One example of an energy harvesting system in shown in FIG. **7**, which is a cross-sectional view of a container closure system **70** including a stopper **700** disposed within a cartridge **790** and powered by a wireless system **780**. The stopper **700** includes a shell enclosing a medicament **711** in the cartridge **790** and, in some instances, the shell **701** includes a plurality of sealing elements engaged to an inner surface of the housing **701**. The stopper **700** also includes an insert **710** having an embedded electronics assembly **771**, **778**, **779**. In operation, the electronics assembly **771**, **778**, **779** is, in some embodiments, integrated with or carried by the insert **710**, whereby inserting the insert **710** into the shell **701** of the stopper **700** includes inserting the electronics assembly **771**, **778**, **779** into a cavity formed in the shell **701**. The electronics assembly **771**, **778**, **779** includes a sensing device(S) **779**, a wireless device (WL) **778**, and a capacitive device (CE) **771**.

[0135] In operation, the sensing device **779** is configured to sense the position of the stopper **700** within the cartridge

790, and the wireless device **778** is configured to communicate with an external electronic device (not shown) in order to communicate information from the sensor device **779**. The capacitive device **771** is configured to provide electric power to the sensing device **779** and the wireless device **778** by way of wireless inductive charging from a wireless signal **781** located in proximity to the cartridge **790**. In some embodiments, the capacitive device **771** includes capacitive circuitry that is configured to receive power wirelessly from, for example, a smartphone **780** via a nearfield communication protocol (NFC) signal **781**, or by a typical wireless charging device with other means of inductive loading, in order to provide enough energy for initiating and performing measurements with the sensing device **779** in the cartridge **790** and for transmitting back the results using the wireless device **778**.

[0136] Another example of an energy harvesting system is the use of a piezo technology to collect energy from the mechanical forces occurring in between the stopper and plunger during, for example, injector handling or an injection operation, to provide enough energy for initiating and performing the measurement in the cartridge and for transmitting back the results. FIGS. **8**A and **8**B are cross-sectional views of a container closure system **80** including a stopper **800** disposed within a cartridge **890** and the stopper **800** includes a sensor insert **810** having electronics powered by a piezoelectric system. FIG. **8**A shows the container closure system **80** being acted upon by a plunger **860** advancing (e.g., along arrow **897**) to contact the insert **810** deposed in the stopper **800**. The stopper **800** encloses a medicament **811** in the cartridge **890** where a shell **801** of the stopper **800** is sealingly engaged to an inner surface of the cartridge **890**. The insert **810** is disposed in a cavity of the shell of the stopper **800**, and the insert **810** includes an embedded electronics assembly **872**, **878**, **879**. The insert **810** is configured to at least partially deflect under the force of the plunger **860** or otherwise enable a transfer for force from the plunger to a portion of the electronics assembly **872**, **878**, **879**. In operation, the electronics assembly **872**, **878**, **879** is, in some embodiments, integrated with or carried by the insert **810**, whereby securing the insert **810** to the stopper **802** includes inserting the electronics assembly **872**, **878**, **879** into the cavity of the shell of the stopper **800**. The electronics assembly **872**, **878**, **879** includes a sensing device(S) **879**, a wireless device (WL) **878**, and a piezoelectric element (PE) **872**.

[0137] In operation, the sensing device **879** is configured to sense the position of the stopper **800** within the cartridge **890**, and the wireless device **878** is configured to communicate with an external electronic device (not shown) in order to communicate information from the sensor device **879**. The piezoelectric element **872** is configured to provide electric power to the sensing device **879** and the wireless device **878** by way of transforming a portion of the force applied to the stopper **800** into electric energy. As shown in FIG. **8**B, the piezoelectric element **872** is transformed from a position **898** of FIG. **8**A to a deflected position **899** of FIG. **8**B by the force applied to the insert **810** by the plunger **816** (during the motion indicated by arrow **897**). The transformation of the piezoelectric element **872** from position **898** to the deflected position **899** absorbs energy and converts a portion of it to electric energy.

[0138] Another example of an integrated energy harvesting device is the inclusion of Peltier elements (PE) to convert the temperature differences between refrigeration (e.g., of the pen or injector during storage) and warming (e.g., exposure to room temperatures) into electric energy to provide enough energy for initiating and performing the measurement in a cartridge of the injector/pen and for transmitting back the results. FIGS. **9**A and **9**B are cross-sectional views of a container closure system **90** including a stopper **900** disposed within a cartridge **990**, where the stopper **900** includes a Peltier thermoelectric device powering internal electronics devices. FIG. **9**A shows a cartridge **990** being stored in a low temperature environment (e.g., 4° C.) and having a stopper **900** disposed in the cartridge **990** and plunger **960** positioned against an insert **910** disposed in a shell **901** of the stopper **900**. The shell **901** of the stopper **900** encloses contain a medicament **911** in the cartridge **990** by being sealingly engaged to an inner wall of the cartridge **990**. The insert **910** includes an electronics assembly **973**, **978**, **979**, and the insert **910** or shell **901** is configured to at least partially transfer thermal energy to the electronics assembly **973**, **978**, **979**. The electronics assembly **973**, **978**, **979** includes a sensing device(S) **979**, a wireless device (WL) **978**, and a thermoelectric element (TE) **973**.

[0139] In operation, the sensing device **979** is configured to sense the position of the stopper **900** within the cartridge **990**, and the wireless device **978** is configured to communicate with an external electronic device (not shown) in order to communicate information from the sensor device **979**. The thermoelectric element **973** is configured to provide electric power to the sensing device **979** and the wireless device **978** by way of generating energy when the temperature of the thermoelectric element changes. As shown in FIG. 9B, the container closure system **90** is moved to a relatively higher temperature environment (e.g., 20° C.) and the thermoelectric element **973** is heated by absorbing thermal energy from the environment outside the container closure system **90** (during the temperature transition indicated by arrow **999**). The absorption of thermal energy by the thermoelectric element **973** generates electric energy to power, for example, the sensing device **979** and the wireless device **978**.

[0140] FIGS. **7-9** generally show inserts **710**, **810**, **910** of a generic shape in order to illustrated the embedded electronics components, but it is understand that inserts **710**, **810**, **910** may, in some instances, have shapes that include any of the stepped profile described herein.

[0141] FIG. **10** is a side view of a stopper **1000** showing the outline of an insert **1010** disposed in a cavity of a shell **1001** of the stopper **1000**, where the insert **1010** has a straight stepped profile. The insert **1010** is disposed in the shell **1001** such that a proximal end **1011** of the insert **1010** is below the proximal end **1002** of the shell **1002**, such that the insert **1010** is able to be hermetically sealed in the shell **1001** at the proximal end **1002**. The hermetic seal protects the insert **1010** from contact with processing aids during washing and sterilization of the assembled stopper **1000**. Further, the hermetic seal protects other stoppers in bulk shipping from contamination if e.g. a battery is leaking. Stoppers with a hermetically sealed insert can be easily washed, sterilized and thereafter processed in aseptic manufacturing areas. The shell **1001** includes a distal end **1003** and a proximal end **1002**, with the cavity (not visible) formed in the proximal end **1002**, where the insert **1010** is shown occupying the volume of the cavity. The insert includes a proximal end **1011** flush with the proximal end of the shell **1001**, and a distal end **1012** located close to the distal end of the shell **1001**. The insert **1010** defines straight (e.g., longitudinal) exterior surface segments **1014***a-c*, where discontinuities in between each of the exterior surface segments **1014***a-c* create stepped surfaces **1015***a*, **1015***b* between the exterior surface segments **1014***a-c*. The stepped surfaces **1015***a*, **1015***b* and the exterior surfaces **1014***a-c* (and, for completeness, the distal end **1012**) are mirrored by the interior surface of the cavity of the shell **1001** in order to interface the insert **1010** in the cavity of the shell **1001** as closely as possible. In this manner, the stepped surfaces **1015***a*, **1015***b* (illustrated as horizontal surfaces) of the insert **1010** are abutting corresponding interface surfaces of the cavity, and, when a force (e.g., from a plunger rod) is applied to the proximal end **1011** of the insert **1010**, a portion of the force is transferred to insert across the interface surfaces abutting the stepped surfaces **1015***a*, **1015***b*.

[0142] FIG. **11** is a side view of a stopper **1100** showing the outline of an insert **1110** in a cavity of a shell **1101** of the stopper **1100**, where the cavity and the insert **1110** define acute angled projections that inwardly stepped surfaces **1116** configured to radially retain the shell **1101** to the insert **1110**. The shell **1101** includes a distal end **1103** and a proximal end **1102**, with the cavity (not visible) formed in the proximal end **1102**, where the insert **1110** is shown occupying the volume of the cavity. The insert includes a proximal end **1111** being flush with the proximal end of the shell **1101**, and a distal end **1112** located close to the distal end of the shell **1101**. The insert **1110** defines straight (e.g., longitudinal) exterior surface segments **1114***a-c*, where discontinuities in between each of the exterior surface segments **1114***a-c* create inwardly stepped surfaces **1116***a*, **1116***b* between the exterior surface segments **1114***a-c*. The inwardly stepped surfaces **1116***a*, **1116***b* and the exterior surfaces **1114***a-c* (and, for completeness, the distal end **1112**) are mirrored by the interior surface of the cavity of the shell **1101** in order to interface the insert **1110** in the cavity of the shell **1101** as closely as possible. In this manner, the inwardly stepped surfaces **1116***a*, **1116***b* of the insert **1110** are abutting corresponding interface surfaces **1106***a*, **1106***b* of the cavity, and, when a force (e.g., from a plunger rod) is applied to the proximal end **1111** of the insert **1110**, a portion of the force is transferred to insert across the interface surfaces **1116***a*, **1116***b* abutting the corresponding inwardly stepped surfaces **1106***a*, **1106***b*. In addition, when the shell **1101** is constructed from an elastomeric material, the inwardly stepped surfaces **1116***a*, **1116***b* prevent radial (outward) deflection of the shell **1101** away from the insert **1110**.

[0143] FIGS. **12** and FIG. **13** illustrate stoppers **1200**, **1300** corresponding to the configurations of FIGS. **10** and **11** but with inserts **1210**, **1310** having conical tapering exterior surfaces **1214***a-c* (FIGS. **12**), and **1314***a-c* (FIG. **13**) forming the stepped surfaces **1215***a*, *b* and **1316***a*, *b*. In FIG. **13**, the sloped interface surfaces **1316***a*, **1316***b* reduce the induced radial (outward) component of the applied force that is created by the interaction of the conical exterior surfaces **1314***a-c* and the cavity.

[0144] FIG. **14** is a side view of a stopper **1400** showing the outline of an insert **1410** disposed in a cavity of a shell **1401** of the stopper **1400**, where the insert **1410** is secured in the cavity with snap-fit features **1417**, **1418**, **1419**. The insert **1410** defines an exterior surface **1414***a-c* of a straight (i.e., longitudinal) profile, and the proximal an exterior surface **1414***a* and the middle exterior surface **1414***b* have formed therein three grooves **1417**, **1418**, **1419** configured to mate with the snap-fit elements **1407**, **1408** (e.g., circular or elliptical ring sections protruding into the cavity of the shell **1401**) and retain the insert **1410** in the shell **1401**. In operation, the insertion of the insert **1410** in the shell **1401** deflects or otherwise deforms the snap-fit elements **1407**, **1408**, **1409**, which, in some instances, are constructed from an elastomeric material, until the snap-fit elements **1407**, **1408**, **1409** align with a corresponding groove **1417**, **1418**, **1419** in the insert **1410**. When aligned, the snap-fit elements **1407**, **1408**, **1409** relax radially inward into the grooves **1417**, **1418**, **1419** of the insert **1410**, and prevent easy removal of the insert **1410** from the shell **1401**.

[0145] FIG. **15** is a side view of a stopper showing the outline of a larger insert **1510** disposed in a shell **1501** of a stopper **1500**, where the insert **1510** is fully inserted into the stopper **1500**. The insert **1510** of FIG. **15** includes a larger distal end **1512** (i.e., as compared the inserts illustrated in the previous figures) such that the insert **1510** has a generally cylindrical shape from a proximal end **1511** to the distal end **1512**. In this configuration, the insert **1510** is able to house a larger electronics assembly, which is able to have a larger portion at the distal end **1512** of the insert **1510** as compared with tapered shapes. The insert **1510** includes exterior surfaces **1514***a-c*, with a recessed exterior surface **1514***b* positioned between a proximal exterior surface **1514***a* and a distal exterior surface **1514***c*, each having a larger diameter than the recessed exterior surface **1514***b*. A stepped surface **1515** and an angled surface **1517** connect the recessed exterior surface **1514***b* with the adjacent surfaces **1514***a*, **1514***c*. In this way, the insert **1510** includes a combination of both a step design and a snap-fit design. The step design is present in the stepped surface **1515** between the proximal exterior surface **1514***a* and the recessed exterior surface **1514***b*, where the stepped surface **1515** configured to deliver a force to the shell **1501** as detailed above (see, for example, FIG. **3**). Further, the recessed exterior surface **1514***b* and the adjacent stepped surface **1515** and angled surface **1517** define a groove for a snap-fit region **1508** of the shell **1501** to occupy and secure the insert **1510** to the shell **1505**.

[0146] FIG. **16**A is a side view of a stopper **1600** showing the outline of an insert **1610** disposed in a shell **1601** of the stopper **1600**. The configuration of FIG. **16**A is similar to that of FIG. **15**, except that the proximal end **1611** of the insert **1610** is flush with the proximal end **1602** of the shell **1601**. Because of the larger distal end thickness of the insert **1610** some examples include different constructions of the shell **1601** to reduce the possibility of air being trapped between the distal end **1612** of the insert **1610** and the shell **1601** after insertion, which can negatively affect the performance of the stopper **1600** disposed in a container or syringe by reducing the strength of the coupling between the insert **1610** and the shell **1601**, which, in some instances, reduces the accuracy of the position of the shell **1601** in a container with respect to the position of a plunger rod contacting the insert **1610**.

[0147] FIG. **16**B is a side view of the stopper of FIG. **16**A modified to have a concave arch section **1607** forming a portion of the distal end **1603** of the shell **1601**. This feature ensures that during assembly of the insert **1610** entrapped air is pushed away from the front end when the insert **1610** makes first contact to convex distal end **1606** of the cavity. The convex distal end **1606** will be pushed out by the insert to lay flat against the distal end **1612** of the insert **1610**. With insertion of the insert **1610**, the insert **1610** makes immediate contact with and deflects the convex distal end **1606** such that the concave arch section **1607** is deflected to form a flat exterior surface of the distal end **1603** and thus forming a designed shape of the stopper **1600**. By providing that the insert **1610** makes contact with the convex distal end **1606**, sensor functionality is improved, because it ensures a reliable location relationship between the sensor (e.g., in electronic device **680** of FIG. **6**) in the insert **1610** and the stopper shell **1601** and, therefore, a reliable relationship between the sensor and any drug product filled into the cartridge outside the distal end **1603** of the stopper **1600**. This improves the precision and predictability of sending and received a sensing signal though the distal end **1603** of the stopper **1600**. In some instances, the exterior surface of the distal end **1603** is deflected to a non-flat shape upon insertion of the insert **1610**, for example, a rounded surface.

[0148] FIG. **17** is a side view of a stopper **1700** showing the outline of an insert **1710** disposed in a cavity in a shell **1701** of the stopper **1700**, where the insert **1710** is sized to contain a larger electronics assembly and is secured to the shell **1701** using two snap-fit elements. The insert **1710** includes two recessed exterior surfaces **1714***a*, *b*, which define two stepped elements **1715***a*, *b*, and two angled surfaces **1517**, which are, together, configured to receive two corresponding snap-fit portions **1708***a*, *b* of the inner wall of the cavity of the shell **1701**. In operation, the stepped elements **1715***a*, *b* distribute a portion of a force applied to the stopper **1700** to the shell and resist movement of a distal end **1712** of the insert **1710** into the cavity beyond a designed position with respect to a distal end **1703** of the shell **1701**. During assembly, the snap-fit portions **1708***a*, *b* deflect to allow the insert **1710** to pass into the cavity of the shell **1701** and relax against the recessed exterior surfaces **1714***a*, *b* of the insert **1710** once the insert reaches its intended positioned in the shell **1701**. In some instances, more than two snap-fit elements are used to secure the insert **1710** in the shell **1701**.

[0149] FIGS. **18**A and **18**B are cross-sectional and top views, respectively of a stopper **1800** with an insert **1810** disposed in a cavity **1806** (shown in FIG. **18**B of a shell **1801** of the stopper **1800**, where the cavity **1806** includes a vent channel **1820**. In operation, when the insert **1810** is inserted into the cavity **1806** air trapped between the distal end **1812** of the insert **1810** and the cavity **1806** is directed out of the stopper **1800** by passing through the vents **1820** to a space **1821** between the proximal end **1811** of the insert **1810** and the proximal end **1802** of the shell **1801** at the proximal end of the stopper **1800**. FIG. **18**C is a cross-sectional view of the stopper of FIG. **18**A showing vent channels **1820** though a middle portion of the shell **1801**.

[0150] FIG. **19**A is a cross-sectional view of a stopper **1900** with a conical cavity **1906** and an arched distal end **1903**. The conical cavity **1906** is configured to accept a conical insert (**1910** of FIG. **19**B) and the arched distal end **1903** deflects to enable the insert **1910** to be fully inserted into the shell **1901**. The arched distal end **1903** is flanked by vent channels **1920** in walls of the shell **1901**, and the vent channels **1920** provide an egress for any air that may be trapped between the insert **1910** and the shell **1901** during insertion. In some instances, the vent channels **1920** do not extend fully to the open end of the cavity **1906**, but instead provide small voids at the sides of the closed end of the cavity **1906** to enable to insert **1910** to be fully inserted without trapping air between the distal end of the insert **1910** and the shell **1901**. In this way, the vent channels **1920** allow the distal end of the insert **1910** to contact the arched distal end **1903** and deflect the arched distal end **1903** during insertion until the distal end of the insert **1910** rests flush against the deflected arched distal end **1903**, as shown in FIG. **19**B. Also, the wall of the shell **1901** at the open end of the cavity may act as a flap valve against the outer wall of the insert **1910** to prevent ingress of expelled air back into space between the shell **1901** and the insert **1910**.

[0151] FIG. **19**B is a side view of the stopper **1900** of FIG. **19**A showing the outline of the insert **1910** disposed in the stopper **1900**, where the arched distal end **1903** is shown in a deflected position. In operation, the inwardly tapering sidewalls of the cavity **1906** deflect radially outward slightly to accept the insert **1910** when the arched distal end **1903** is deflected, but the walls maintain an inward taper to accept the conical shape of the insert **1910** and resist over-insertion of the insert **1910** into the cavity **1906** by being pressed against the walls of a container (not shown) in which the stopper **1900** is disposed and by the material properties of the arched distal end **1903** resisting a tensioned deformation as the insert **1910** pushes radially outward under a force from a plunger rod. While FIGS. **16**, **19**A, and **19**B show a shell **1601**, **1901** without step features, another embodiment of the present disclosure is a shell with an arch distal region and step features (of FIGS. **3**-**6**C and **10**-**15**) where the step features, for example, prevent over insertion of an insert into the cavity and/or retain the shell walls with sloped stepped features (where the sloped stepped features are shown in FIGS. **5** and **11**). Any one or combination of the step features shown in FIGS. **3**-**6**C and **10**-**15** are compatible with the arch configurations of FIGS. **16**, **19**A, and **19**B.

[0152] FIG. **20** is a cross-section illustration of a stopper **2000** disposed in a container **2091**. The stopper **2000** includes a rigid shell **2001** with in-molded seals **2004**, **2005** and an insert **2010** secured inside a cavity of the rigid shell **2001** with an adhesive element **2060**. The in-molded seals **2004**, **2005** are arranged on the exterior surface of the rigid shell **2001** to create a sealing interface between the stopper **2000** and an inner surface of the container **2091**. As an alternative to in-molded seals, also O-ring seals may be used (not shown). The stopper **2000** also includes a back-end closure cap **2080** inserted into the cavity behind the insert **2010** in order to seal the insert **2010** into the cavity of the stopper **2000**. The adhesive element **2060** is arranged in the cavity of the rigid shell **2001** at the closed end such that the insert **2010** contacts the adhesive element **2060** when maximally inserted into the cavity. The adhesive element **2060**, in some instances, is made from an adhesive material or includes an adhesive material on the exterior surfaces. The adhesive element **2060** is contacted by the insert **2010** when the insert **2010** is inserted into the cavity and affixes the insert **2010** to the closed end of the cavity of the rigid shell **2001**. In some instances the adhesive element **2060** is contacted by the insert **2010** when the back-end closure cap **2080** is inserted into the cavity and contacts the insert **2010** such that the insert **2010** is driven forward against the adhesive element **2060** in order for the back-end closure cap **2080** to reach a position in the cavity where the back-end closure cap **2080** sealingly engages the rigid shell **2001**. In some instances, the cavity of the rigid shell **2001** defines a front (i.e., distal) end **2006** defining a shape that corresponds to a shape of the distal end **2012** of the insert **2010**. In some instances, the shape of the distal end **2006** of the cavity of the rigid shell **2001** is defined by the adhesive element **2060** extending into the cavity. In some instances, the adhesive element **2060** is attached to the insert **2010** prior to assembly with the rigid shell **2001**.

[0153] FIG. **21** is a cross-section illustration of a stopper **2100** disposed in a container **2191**. The stopper **2100** includes a rigid shell **2101** with in-molded seals **2104**, **2105** and an insert **2110** pressed into a deformable element **2170** in a cavity of the rigid shell **2101**. The stopper **2100** also includes a back-end closure cap **2180** inserted into the cavity behind the insert **2110** in order to seal the insert **2110** into the cavity of the stopper **2100**. The deformable element **2170** is arranged in the cavity of the rigid shell **2101** at the closed end such that the insert **2110** contacts the deformable element **2170** when maximally inserted into the cavity. Alternatively, the deformable element **2170** may be attached to the insert **2110** prior to assembly into the rigid shell **2101**. The deformable element **2170** is configured to be deformed by the insert upon insertion into the cavity and, in some instances, is made from an elastic and/or plastic deformable material (e.g., silicone, TPE material, glue or wax). In some instances, the deformable element **2170** defines a flat to convex shape made prior to being contacted and deformed by the insert **2110**. The deformable element **2170** is contacted and deformed by the insert **2110** when the insert **2110** is pressed into the cavity and affixes the insert **2110** to the closed end of the cavity of the rigid shell **2101**. In some instances the deformable element **2170** is contacted by the insert **2110** when the back-end closure cap **2180** is inserted into the cavity and contacts the insert **2110** such that the insert **2110** is driven forward against the deformable element **2170** in order for the back-end closure cap **2180** to reach a position in the cavity where the back-end closure cap **2180** sealingly engages the rigid shell **2101**.

[0154] FIG. **22** is cross-section illustration of a stopper **2200** in a container **2291**. The stopper **2200** includes a rigid shell **2201** with in-molded seals **2204**, **2205** and an insert **2210** secured inside a cavity of the rigid shell **2201** with a deformable element **2270**. The stopper **2200** also includes a back-end closure cap **2280** inserted into the cavity behind the insert **2210** in order to seal the insert **2210** into the cavity of the stopper **2200**. The deformable element **2270** is arranged in the cavity of the rigid shell **2201** between the insert **2210** and the back-end closure cap **2280** such when the insert **2210** contacts the closed end of the cavity and the back-end closure cap **2280** is and driven forward against the deformable element **2270** to sealingly engage the rigid shell **2201**, the deformable element **2270** is deformed between the insert **2210** and the back-end closure cap **2280**. In some instances, similar to the arrangement of FIG. **20**, an adhesive element **2260** is arranged in the cavity of the rigid shell **2201** at the closed end such that the insert **2210** contacts the adhesive element **2260** when the deformable element **2270** is pressed between the back-end closure cap **2280** and the insert **2210**. In other instances, the insert **2210** directly engages the closed end of the cavity.

[0155] In some cases, the rigid shell **2001**, **2101**, **2201** is made of more rigid material which is selected to be compatible with the medicament e.g., PP, PE, COC, COP, PTFE or is made of elastomeric material, e.g. butyl rubber, halobutyl rubber, thermoplastic elastomer (TPE), silicone rubbers, polyurethane and the like at least at the distal end **2001**, **2101**, **2201** which is in contact with a medicament in the container **2091**, **2191**, **2291**.

[0156] Those of skill in the art will understand that modifications (additions and/or removals) of various components of the APIs, formulations, apparatuses, methods, systems and embodiments described herein may be made without departing from the full scope and spirit of the present disclosure, which encompass such modifications and any and all equivalents thereof.

[0157] A number of implementations of the present disclosure have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the present disclosure. Accordingly, other implementations are within the scope of the following claims.

What is claimed is:

1. A stopper for use in a cartridge or a syringe of a medical device, the stopper configured to be disposed within a container closure system, the stopper comprising:

a shell comprising a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, the open end defining a cavity and the sidewalls defining an exterior surface sized and shaped to fit inside the container closure system; and

an insert configured to be inserted into the cavity, the insert sized and shaped to receive a force from a plunger rod and distribute the force to the shell to advance the shell into the container closure system,

wherein the closed end of the shell in the cavity defines a convex surface configured to be contacted and deflected by the insert upon insertion of the insert into the cavity,

wherein the closed end of the shell is made of an elastic and/or plastic deformable material; and

wherein the insert comprises an electronic device.

2. The stopper of claim **1**, wherein the closed end of the shell defines an arch region defining the convex surface in the cavity and a concave exterior surface of the closed end of the shell, and wherein the arch region is configured to be deflected by the insert contacting the convex surface during insertion of the insert into the cavity.

3. The stopper of claim **2**, wherein the cavity defines inwardly tapering sidewalls from the open end of the shell to the closed end of the shell, and wherein the inwardly tapering sidewalls are configured to maintain an inward taper when the arch region is deflected by the insert.

4. The stopper of claim **1**, wherein the cavity comprises an interior step element, and wherein the insert comprises a corresponding step element configured to abut the interior step element and distribute at least a portion of the force from the plunger rod to the interior step element.

5. The stopper of claim **4**, wherein the cavity defines a snap-fit feature extending into the cavity, the snap-fit feature configured to retain the insert in the cavity by being deflected or deformed by the insert during insertion of the insert into the cavity until the snap-fit feature relaxes into a corresponding depression in the insert, and wherein an exterior surface of the insert defines the corresponding depression being sized and positioned to accept the snap-fit feature.

6. The stopper of claim **4**, wherein the corresponding step element is sized and positioned to retain the interior step element and resist radial deflection of the interior step element by the force applied to the insert.

7. The stopper of claim **4**, wherein the interior step element defines a first interface surface, and wherein the corresponding step element defines a second interface surface, and wherein the first interface surface abuts against the second interface surface when the insert is inserted into the shell.

8. The stopper of claim **7**, wherein the first interface surface and the second interface surface define acute angles about the closed end of the shell.

9. The stopper of claim **1**, wherein the cavity comprises at least one venting channel extending from the closed end of the shell towards the open end of the shell, the at least one venting channel being sized and positioned to enable air and/or fluid in the cavity to be expelled though the at least one venting channel during insertion of the insert into the cavity.

10. The stopper of any of claim **1**, wherein the insert defines a conically tapering exterior surface configured to enable venting of air in the cavity around an exterior of the insert during insertion of the insert into the cavity.

11. The stopper of claim **1**, wherein the electronic device comprises a sensor configured to generate a sensing signal, wherein the closed end of the shell is configured to allow the sensing signal to pass therethrough.

12. The stopper of claim **11**, wherein the sensor is configured to be responsive to a position of the stopper in the container closure system.

13. The stopper of claim **1**, wherein the electronic device comprises at least one of an energy source, a microcontroller, or a wireless transceiver.

14. The stopper of claim **13**, wherein the wireless transceiver is configured to communicate with an external electronic device.

15. The stopper of claim **14**, wherein the wireless transceiver is an RFID device.

16. The stopper of claim **1**, wherein the electronic device comprises an on-off switch configured to trigger the electronic device by impact on the insert.

17. The stopper according to claim **1**, wherein the electronic device comprises an energy harvesting system.

18. The stopper according to claim **1**, wherein the insert comprises a heat resistive coating, comprising a PTFE coating.

19. A container closure system comprising:

A syringe housing; and

a stopper, comprising:

a shell comprising a closed end and an open end with sidewalls extending between the closed end and the open end along a longitudinal axis of the shell, the open end defining a cavity and the sidewalls defining an exterior surface sized and shaped to fit inside the container closure system; and

an insert configured to be inserted into the cavity, the insert sized and shaped to receive a force from a plunger rod and distribute the force to the shell to advance the shell into the container closure system,

wherein the closed end of the shell in the cavity defines a convex surface configured to be contacted and deflected by the insert upon insertion of the insert into the cavity,

wherein the closed end of the shell is made of an elastic and/or plastic deformable material; and

wherein the insert comprises an electronic device.

wherein the shell is configured to be inserted into the syringe housing prior to the container closure system being filled with a medical product, and the insert is configured to be inserted into the stopper after filling the medical product.

20. The container closure system of claim **19**, wherein the container closure system is a syringe, and wherein the insert is disposed at a distal end of the plunger rod of the syringe and is configured to be inserted into the shell of the stopper after the syringe is assembled into a medical device.

21. The container closure system of claim **19**, wherein the syringe housing is configured for use with one or more of: an autoinjector, a pen injector, or an injection pump.

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